

Indivisible stochastic processes for non-Hermitian and
higher-derivative quantum systems:
detailed balance, broken $\mathcal{PT}$ symmetry, and the explosion of
ghosts

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July 11, 2026 (v1.1: Sec. 10, experimental probes, added)

Abstract

Barandes’ stochastic–quantum correspondence represents unitary quantum theory by *indivisible* stochastic processes on a configuration space, while the antilinearity program of Bender and Mannheim replaces Hermiticity by $\mathcal{PT}$ symmetry, recovering unitarity in a modified inner product. We ask whether these two reconstructions are compatible, and follow the question from the canonical 2×2 $\mathcal{PT}$ dimer to interacting Pais–Uhlenbeck-type ghosts. Our results: (i) unbroken antilinear symmetry is exactly the condition for a probability-conserving indivisible process to exist *without enlarging the configuration space*—on beables generically rotated from the original configurations—and the metric-corrected process is the ordinary process of the isospectral Hermitian partner; (ii) a non-Hermitian Hamiltonian admits a *fixed-beable* representation iff its couplings satisfy a Kolmogorov detailed-balance criterion, and the resulting process is unique—in gauge language, fixed beables exist precisely when the imaginary gauge flux vanishes, making the non-Hermitian skin effect the exact obstruction; (iii) the broken phase, treated as an open subsystem by dilation, supports a valid sub-stochastic process whose indivisibility is *terminal* rather than recurrent—we prove exact closed forms, including the generator floor $-(\sin \theta + \tilde{\kappa})$ (a floor on the indivisibility strength $|L_{12}|$, approached from below by the signed entry)—while late-time information backflow vanishes: the two notions of non-Markovianity dissociate; (iv) the exceptional point is a coordinate singularity of the Hermitian dictionary, not of the process; (v) the free Pais–Uhlenbeck ghost admits a perfectly valid fixed-beable process, whereas the no-ghost quantization exits the sample space; (vi) interacting ghosts realize the probabilist’s classification conservative/transient/explosive, and (vii) a deficiency-index/Feller analysis proves an “explosion dictionary” for the one-dimensional monotone mechanism model: one classical integral decides classical completeness, quantum self-adjointness, and stochastic conservativity, with (viii) a rigorous channel-based multidimensional extension in which conservative completions carry an infinite-dimensional, channel-mixing S -matrix at infinity. For the exact quartic Pais–Uhlenbeck operator we further prove an exact displaced-oscillator identity whose channelization (exact to $O(k^2)$) carries a sign prediction the numerics confirm ($\lambda > 0$: limit-circle channels; $\lambda < 0$: a quantum-completion candidate despite classical blowup), settle deficiency stability at every finite channel number, certify the one-dimensional fiber non-self-adjoint, in a computer-assisted sense, by a glued-tail von Neumann certificate—no compactly supported function can witness deficiency—and conjecturally reduce the remaining $M \rightarrow \infty$ step to Gaussian-beam assembly and error control along numerically characterized

escape rays. The ghost problem, viewed in the stochastic currency, is a recurrence-classification problem, and the minimal explosive process is its only canonical resolution.

Note on provenance. This manuscript is an experiment in AI-conducted research as much as a physics paper. The project began when the author asked the large language model Claude Fable 5 (Anthropic): *“Is it not possible that the antilinear PT /time-reversal symmetry element could maybe predict indivisible non-Markovian processes?”* Everything that followed was performed by the model: the research program, the theorems and proofs, the numerical experiments, the literature search, and this text. The model was repeatedly prompted to attack its own results adversarially and to search for prior work. The author, a professional engineer rather than a physicist, directed and reviewed the sessions but makes no claim to comprehension of the findings, and is circulating the manuscript for independent review by professional physicists. Accordingly, nothing here should be taken on trust: every theorem is stated with a full proof, and every numerical claim is reproducible from the public scripts (see [Code availability](#)), where the originating conversation and project briefing are also preserved. The manuscript is additionally subject to the program’s release gate: repeated adversarial audit passes by fresh AI agents—each independently re-deriving every proof, re-running every script and tracing every quoted number to its output, and verifying every reference against primary records—with circulation requiring three consecutive passes with zero findings; the pass-by-pass ledger of findings and fixes is public in the repository ([AUDIT.md](#) and [histories/](#)).

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1 Introduction

Two contemporary research programs demote the standard Hilbert-space formalism of quantum mechanics to a derived structure, and both elevate the role of time reversal.

The first is Barandes’ *stochastic–quantum correspondence* [1, 2]: quantum theory is equivalent to a class of generalized stochastic processes on a configuration space, in which a definite configuration (a “beable”) exists at all times and the transition matrix $\Gamma(t)$ fails to factorize between measurements. Such processes are *indivisible*: except at special times, no stochastic matrix $\Gamma(t_2, t_1)$ satisfies $\Gamma(t_2) = \Gamma(t_2, t_1)\Gamma(t_1)$. Unitarity, in this reading, is not an axiom but a representation-theoretic consequence.

The second is the antilinearity program of Bender, Mannheim, and collaborators [4, 5, 6]: the correct organizing principle of quantum theory is not Hermiticity but invariance under an antilinear symmetry such as $\mathcal{PT}$. A $\mathcal{PT}$ -symmetric, non-Hermitian Hamiltonian with real spectrum (“unbroken phase”) evolves unitarily with respect to a modified inner product; when eigenvalues form complex-conjugate pairs (“broken phase”), exponentially growing and decaying modes balance in a biorthogonal pairing. The program’s flagship application is the fourth-order Pais–Uhlenbeck (PU) oscillator [9], the quantum-mechanical prototype of conformal (Weyl) gravity [7], whose Ostrogradski ghost is claimed to be exorcised by $\mathcal{PT}$ -symmetric quantization [8].

These programs have, to our knowledge, never been brought into contact (a recent study of divisibility within the correspondence [3] remains within the standard Hermitian/unitary-generated setting and does not treat non-Hermitian or antilinear dynamics). Barandes’ construction rests on ordinary unitarity in a fixed inner product; Mannheim’s unitarity lives in a Hamiltonian-dependent inner product. The natural question is:

Can antilinear ($\mathcal{PT}$) symmetry predict or produce indivisible stochastic processes—and what becomes of the correspondence in the broken phase, at exceptional points, and for the ghosts it was designed to save?

This paper answers the question in stages, each stage exactly solvable or numerically verified, and each feeding the next. Our main results:

1. **Existence gates, but does not generate** (Sec. 2). Unbroken antilinear symmetry is precisely the condition for a valid indivisible process to exist; however, the metric-corrected process is the *ordinary* Barandes process of the isospectral Hermitian partner $h = \rho H \rho^{-1}$ [10]. Closed-system $\mathcal{PT}$ machinery is an isomorphism, not a generalization. Its two costs are dynamics-dependent beables and metric non-uniqueness.
2. **Fixed beables $\Leftrightarrow$ detailed balance** (Sec. 3). The *detailed-balance theorem* (Theorem 3.1): a non-Hermitian H admits a beable-preserving (diagonal-metric) representation iff its couplings satisfy Kolmogorov-type detailed-balance conditions; the metric weights are the inverse stationary distribution, the spectrum is automatically real, and the fixed-beable process is unique—both costs dissolve exactly on this class. In gauge variables the criterion reads: the *imaginary* gauge flux must vanish while real magnetic flux is unconstrained, so the non-Hermitian skin effect [19, 38] is precisely the obstruction to fixed beables (Sec. 3.1).
3. **The broken phase is a valid open process; the two non-Markovianities dissociate** (Sec. 4). Broken- $\mathcal{PT}$ evolution is survival-conditioned dynamics of a contraction; its unitary

dilation defines an ordinary indivisible process whose reduction is sub-stochastic. The *terminal-indivisibility theorem* (Theorem 4.1) gives the generator in closed form: divisibility is lost at t_c exactly when $\det \Gamma$ crosses zero, permanently, with floor $-(\sin \theta + \tilde{\kappa})$; the unbroken phase is recurrent instead. Late-window information backflow, by contrast, vanishes in the broken phase: Barandes indivisibility is the strictly finer probe.

4. **The exceptional point is a coordinate singularity** (Sec. 5). The metric collapses with Petermann scaling $\text{cond}(\eta) \sim \varepsilon^{-1}$ and the lab-time metric-corrected process freezes to the identity; a nontrivial critical process survives only on the rescaled clock $\tau = \kappa t$, in coordinates whose beable map diverges. The open-system description passes through the exceptional point smoothly.
5. **Pais–Uhlenbeck: the two programs pay in different currencies** (Sec. 6). The Hermitian Ostrogradski realization of free PU is a valid, fixed-beable, indivisible process on physical configurations $(x, \dot{x})$ —the unbounded-below energy spectrum is invisible to the process. The $\mathcal{PT}$ /no-ghost quantization is a domain change whose vacuum has no shadow at any truncation: not a rotation of beables but an exit from the sample space.
6. **Interacting ghosts: conservative, transient, explosive** (Sec. 7). Quadratic interactions collapse the instability threshold from the sum to the difference frequency and make the process *transient* (exact cascade laws), yet it remains a valid doubly stochastic process with balanced energy books. Malicious nonlinear interactions with finite-time classical escape make it *explosive*.
7. **The explosion dictionary** (Sec. 8). The *explosion dictionary* (Theorem 8.1, for the one-dimensional monotone mechanism model; Sec. 9 records where it fails beyond that model): a single classical integral—the time of flight to infinity—decides classical completeness, quantum essential self-adjointness (via the WKB identity “deficiency norm $\approx$ time of flight”), and stochastic conservativity (Feller). Conservative completions reflect probability off infinity at the classical round-trip time; the canonical object is the minimal, explosive, sub-stochastic process.
8. **Multidimensional channel theory** (Sec. 9). Since classical incompleteness and quantum non-self-adjointness can come apart [26], we prove channel-based statements: quadratic ghosts are rigorously safe (Nelson’s theorem); separable, central, and boundedly-coupled malicious systems have deficiency indices (∞, ∞) , so conservative completions carry a channel-mixing *S-matrix at infinity*. The essential self-adjointness of the exact quartic-PU operator is posed as a sharply delimited open problem—and then taken as far as computation allows: an exact displaced-oscillator identity, its channelization (exact to $O(k^2)$) carrying a confirmed sign prediction (Proposition 9.7), deficiency stability at every finite channel number (Theorem 9.8) with its $\lambda < 0$ mirror (Corollary 9.9), the no-compact-witness principle (Remark 9.11) together with a genuine glued-tail von Neumann certificate for the one-dimensional fiber, and a Maslov feasibility study conjecturally reducing the remaining $M \rightarrow \infty$ step to Gaussian-beam assembly and error control along the escape rays.

All numerical claims are reproducible from short standalone scripts (see [Code availability](#)). Sections 3, 4, 8 and 9 contain the theorems; readers interested primarily in the ghost problem may read Secs. 6–9 independently after the preliminaries.

2 Preliminaries and the deflationary observation

2.1 The stochastic–quantum correspondence

Following [1, 2], a *generalized stochastic system* on a configuration set $\{1, \dots, n\}$ is specified by a family of column-stochastic matrices $\Gamma(t)$, $\Gamma_{ij}(t) \geq 0$, $\sum_i \Gamma_{ij}(t) = 1$, giving the probability of configuration i at time t given j at time 0. The system is *divisible* on (t_1, t_2) if there is a stochastic matrix $\Gamma(t_2, t_1)$ with $\Gamma(t_2) = \Gamma(t_2, t_1) \Gamma(t_1)$, and *indivisible* if such factorizations fail except at isolated times (“division events”). The stochastic–quantum dictionary represents

$$\Gamma(t) = \Theta(t) \odot \Theta^*(t), \quad \Theta(t) = U(t) = e^{-iHt}, \quad \Gamma_{ij} = |U_{ij}|^2, \quad (1)$$

with $\odot$ the entrywise product; $|U|^2$ is doubly stochastic if U is unitary (the converse fails: stochasticity of $|U|^2$ fixes only the moduli), and indivisibility expresses interference, $|UV|^2 \neq |U|^2|V|^2$. Two facts matter below: the family $\Gamma(t)$ need not compose (indivisibility *is* its failure to compose), and only per-time unistochasticity is required.

If $\Gamma(t_1)$ is invertible, the candidate divisor is unique,

$$M = \Gamma(t_2) \Gamma(t_1)^{-1}, \quad (2)$$

and has unit column sums automatically; divisibility on (t_1, t_2) holds iff M is entrywise nonnegative (with column sums ≤ 1 in the sub-stochastic case). Infinitesimally, with the generator $L(t) = \dot{\Gamma}(t) \Gamma(t)^{-1}$, divisibility across t (classical P-divisibility) holds iff the off-diagonal entries of $L(t)$ are nonnegative.

2.2 Pseudo-Hermitian and $\mathcal{PT}$ -symmetric quantum mechanics

A diagonalizable H with antilinear symmetry has eigenvalues either real or in complex-conjugate pairs [5, 6]. If the spectrum is real and a positive-definite metric η exists with

$$H^\dagger \eta = \eta H, \quad (3)$$

then $\rho = \eta^{1/2}$ furnishes a Hermitian partner $h = \rho H \rho^{-1}$ and a unitary $u(t) = \rho e^{-iHt} \rho^{-1}$ [10]. The canonical biorthogonal choice is $\eta = (VV^\dagger)^{-1}$, V the right-eigenvector matrix. Our laboratory is the canonical dimer

$$H(\theta, s) = \begin{pmatrix} e^{i\theta} & s \\ s & e^{-i\theta} \end{pmatrix}, \quad E_\pm = \cos \theta \pm \sqrt{s^2 - \sin^2 \theta}, \quad (4)$$

which is $\mathcal{PT}$ -symmetric ($PH^*P = H$, $P = \sigma_x$): unbroken for $s > \sin \theta$, broken for $s < \sin \theta$, with an exceptional point (EP) at $s_* = \sin \theta$.

2.3 Existence gates; deflation; two costs

Throughout we use $\theta = \pi/4$ and, unless stated, $t = 1.3$.

Proposition 2.1 (metric-corrected process; deflation). *In the unbroken phase, $\Gamma(t) = |\rho e^{-iHt} \rho^{-1}|^2$ is doubly stochastic and generically indivisible; in the broken phase no positive-definite solution of (3) exists and no such process exists. Moreover $\rho e^{-iHt} \rho^{-1} = e^{-iht}$: the metric-corrected process is the ordinary Barandes process of the Hermitian partner h .*

Proof. If $\eta = \rho^2$ solves (3) with ρ positive definite, then $h := \rho H \rho^{-1}$ obeys $h^\dagger = \rho^{-1} H^\dagger \rho = \rho^{-1} (\eta H \eta^{-1}) \rho = h$, so h is Hermitian, $\rho e^{-iHt} \rho^{-1} = e^{-iht}$ is unitary, and the entrywise moduli-squared of a unitary are doubly stochastic (unistochastic). Indivisibility at generic t is exhibited numerically below. In the broken phase the spectrum contains a complex-conjugate pair; similarity to a Hermitian h would force a real spectrum, so no positive-definite solution of (3)—and hence no process of this form—exists. $\square$

Numerically, at $s = 1$ the naive column sums of $|U|^2$ are (3.23, 1.30) (here and throughout, $\|\cdot\|$ on matrices is the Frobenius norm), while the corrected process is $\Gamma = \begin{pmatrix} 0.368 & 0.632 \\ 0.632 & 0.368 \end{pmatrix}$ with $\|\Gamma(2t) - \Gamma(t)^2\| \approx 0.93$; at $s = 0.5$ (spectrum $0.7071 \pm 0.5i$) every Hermitian intertwiner is indefinite and $\|e^{-iHt}\|$ grows as 1.58, 2.04, 3.61, 10.4, 77.2 at $t = 0.5, 1, 2, 4, 8$. Thus unbroken antilinearity is exactly the *gatekeeper* for process existence, while the process itself is isomorphic to an ordinary one—closed-system $\mathcal{PT}$ machinery cannot produce a new class of indivisible processes. The isomorphism carries two costs: (a) ρ is generically non-diagonal, so the beables rotate, and the rotation depends on H ; (b) the metric, hence the process, is non-unique. Both costs are resolved exactly in the next section.

3 Fixed beables and Kolmogorov detailed balance

We ask which H admit a metric diagonal in the configuration basis, so that the beables are preserved.

Theorem 3.1 (fixed-beable characterization). *A matrix H admits a positive diagonal metric $\eta = \text{diag}(d)$, $d_i > 0$, satisfying (3) iff all of the following hold:*

- (K1) *every diagonal entry H_{ii} is real;*
- (K2) $H_{ij} = 0 \Leftrightarrow H_{ji} = 0$;
- (K3) $H_{ij}H_{ji} > 0$ for every coupled pair $i \neq j$;
- (K4) (Kolmogorov cycle condition) *around every cycle of the coupling graph, the product of the moduli $|H_{ij}|$ taken one way equals the product taken the other way.*

When they hold, d is unique up to one positive scale per connected component of the coupling graph.

Proof. The component form of (3) with $\eta = \text{diag}(d)$ is $d_i H_{ij} = d_j \overline{H_{ji}}$. The $i = j$ component gives (K1); positivity of d forces (K2); the phase of the off-diagonal relation gives (K3); its modulus gives $d_i/d_j = |H_{ji}|/|H_{ij}|$, whose consistency around closed loops is (K4). Conversely, define d along a spanning forest; (K4) makes it well defined, and positivity is automatic. $\square$

Corollary 3.2. *Under (K1)–(K4): (i) $h = \eta^{1/2} H \eta^{-1/2}$ is Hermitian, so the spectrum of H is automatically real—unbroken antilinearity is derived, not assumed; (ii) with $\rho = \text{diag}(\sqrt{d})$ the process is the pure reweighting $\Gamma_{ij}(t) = (d_i/d_j) |U_{ij}(t)|^2$, invariant under the residual scale freedom, hence unique; (iii) with classical hopping rates $q(i \rightarrow j) := |H_{ji}|$, the distribution $\pi_i \propto 1/d_i$ satisfies detailed balance $\pi_i q(i \rightarrow j) = \pi_j q(j \rightarrow i)$: the metric weights are the inverse Kolmogorov equilibrium distribution [17, 18]; (iv) the fixed-beable class is exactly the positive-diagonal similarity orbit of the Hermitian matrices (“detailed-balanced gain/loss”).*

Three instructive cases (all verified numerically, including 200/200 randomized trials at $n = 3, \dots, 6$ in which an independent null-space solver reproduces the constructive d):

(a) $n = 2$. Cycles beyond pairs are absent, so (K4) is vacuous: fixed-beable $\Leftrightarrow H_{11}, H_{22} \in \mathbb{R}$ and $H_{12}H_{21} > 0$ (or $H_{12} = H_{21} = 0$, the decoupled case). The dimer (4) fails (K1) for every $\theta \neq 0, \pi$: within that family, fixed beables $\Leftrightarrow$ Hermiticity, consistent with the rotation found in Sec. 2. The cycle condition first bites at $n \geq 3$.

(b) *A genuinely non-Hermitian fixed-beable 3×3 example.* Take $d = (1, 2, 4)$, real diagonal $(0.3, -0.4, 1.0)$, upper entries $H_{12} = 1 + i$, $H_{13} = 0.8i$, $H_{23} = 1.5 e^{i\pi/5}$ and lower entries $H_{ji} = (d_i/d_j)\overline{H_{ij}}$. Then $\|H - H^\dagger\| = 1.69$, yet the spectrum is real $\{-1.319, 0.210, 2.009\}$; the naive column sums $(0.40, 1.17, 3.04)$ are repaired to a doubly stochastic Γ with $\|\Gamma(2t) - \Gamma(t)^2\| \approx 0.41$ by the reweighting alone—no beable rotation.

(c) *Only (K4) fails.* $H = \text{diag}(1, 2, 3)$ plus the asymmetric positive ring with $(H_{12}, H_{23}, H_{31}) = (0.6, 0.6, 0.6)$ and $(H_{21}, H_{32}, H_{13}) = (0.3, 0.3, 0.3)$: cycle products $0.216 \neq 0.027$. The spectrum is still real $\{0.847, 1.840, 3.313\}$, a positive-definite (non-diagonal) metric exists and yields a valid indivisible process, but the diagonal-intertwiner solution space is exactly zero and ρ has an 18% off-diagonal fraction: *(K4) gates fixed beables specifically, not process existence.*

Remark 3.3 (relation to prior work). The matrix-algebra core of Theorem 3.1 is classical: diagonal symmetrizability of nonnegative matrices was characterized by Parter and Youngs [27], and cyclic-product criteria for diagonal similarity by Engel and Schneider [28]. The contribution here is the identification of this class with *beable preservation* in the stochastic–quantum correspondence, the detailed-balance dictionary for the metric, the automatic reality of the spectrum in the pseudo-Hermitian setting, and the uniqueness of the resulting process. In the reverse direction—non-Hermitian quantum techniques applied to classical Markov processes violating detailed balance—see [29].

3.1 Gauge form: the non-Hermitian skin effect is the obstruction

Conditions (K2)–(K3) have a gauge-theoretic meaning: they hold iff the couplings can be written *uniquely* as

$$H_{jk} = t_{jk} e^{i\varphi_{jk} + w_{jk}}, \quad t_{jk} = t_{kj} > 0, \quad \varphi_{jk} = -\varphi_{kj}, \quad w_{jk} = -w_{kj}, \quad (5)$$

i.e. as a lattice Peierls substitution $p \rightarrow p - A - iW$ with a *real* gauge field φ (link variables of A) and an *imaginary* gauge field w (link variables of W , the Hatano–Nelson field [19]); (K1) is the absence of on-site gain/loss. In these variables the detailed-balance theorem (Theorem 3.1) becomes:

Theorem 3.4 (skin obstruction). *For a gauge-form Hamiltonian (5) with real on-site potential:*

- (i) *a fixed-beable representation exists iff the imaginary flux vanishes through every cycle, $\sum_C w = 0$; then $w_{jk} = \chi_j - \chi_k$ is pure gauge, the metric is the exponentiated imaginary gauge transformation $\eta = \text{diag}(e^{-2\chi})$, and $\Gamma_{jk}(t) = e^{2(\chi_k - \chi_j)} |U_{jk}(t)|^2$. The real flux (magnetic field) is unconstrained.*
- (ii) *With classical rates $q(j \rightarrow k) = |H_{kj}|$, the imaginary flux is half the cycle affinity of the rate graph: it vanishes iff the classical chain is reversible (Kolmogorov), and nonzero imaginary flux is equivalent to a nonzero steady-state cycle current.*

(iii) (Hatano–Nelson) A ring with uniform w carries flux Nw : no fixed beables exist. For the uniform ring (uniform t , uniform w , constant on-site potential) the periodic-boundary spectrum is moreover a complex loop (the spectral winding that classifies the skin effect [38]), so there no closed process exists at all; a non-uniform ring with nonzero net flux still has no fixed beables, but its flux can be too weak to break spectral reality, leaving a closed rotated-beable process possible (the diagonally dominated triangle below is exactly such a case). An open chain is a tree: fixed beables exist at every finite N with weights $d_i = e^{2wi}$; the boundary-condition dichotomy of the skin effect is the flux/pure-gauge dichotomy.

Proof. (i) restates (K4) in the variables (5): cycle products of moduli agree in the two orientations iff $e^{\sum_C w} = e^{-\sum_C w}$. (ii) is Kolmogorov’s criterion applied to q , whose cycle ratios are $e^{2\sum_C w}$. (iii) a ring has one independent cycle; a tree has none. $\square$

Numerically (Fig. 1): over 50 random gauge-form graphs with random *real* fluxes and pure-gauge w , the constructed metric is $e^{-2\chi}$ exactly and the spectrum real; injecting imaginary flux through any one cycle collapses the diagonal-intertwiner space to zero. For the Hatano–Nelson chain the OBC skin eigenvectors equal the gauge factor e^{-wi} times the Hermitian-chain eigenvectors to 5×10^{-15} , and $\text{cond}(\eta) = e^{2w(N-1)}$ (an analytic identity, verified to four digits)—so the fixed-beable representation, while exact at every finite N , degenerates exponentially in the thermodynamic limit, reproducing the unbounded-metric pathology of Sec. 6 in one-dimensional lattice form. On the classical side, the steady-state ring current vanishes (to 10^{-17}) exactly when the imaginary flux does. Finally, in a regime where the flux is too weak to break spectral reality (a diagonally dominated triangle), the binary criterion trips at any nonzero flux while the beable rotation grows continuously with it: the obstruction is quantitative below the antilinear transition.

The non-Hermitian skin effect—spectral winding under periodic boundaries, boundary pileup under open ones—is thus exactly the obstruction to a beable-preserving stochastic representation, and equals detailed-balance violation (irreversibility) of the underlying classical rate graph.

Corollary 3.5 (topological classification of the obstruction). *For gauge-form couplings on a graph or lattice, the fixed-beable obstruction is the class of the imaginary gauge field in the first cohomology $H^1(G, \mathbb{R})$: the imaginary field strength must vanish through every contractible cycle (no imaginary magnetic flux), and the imaginary holonomy through every non-contractible one (no winding)—on a d -torus, an $\mathbb{R}^d$ of independent windings, matching the multi-directional classification of skin effects [38], plus the pointwise plaquette condition that band-language winding does not see. The obstruction is finer than net spectral winding: two Hatano–Nelson chains with opposite fields (net winding zero—the doubled structure of the $\mathbb{Z}_2$ skin effect [38]), coupled by reciprocal rungs, are obstructed through their inter-chain cycles of circulation $2w$.*

Numerically (`skin_topology_beables.py`): on a 3×4 torus the diagonal-metric solution space has dimension 1 iff both windings and the plaquette flux vanish, and 0 if any is switched on; on a cylinder, “winding” along the open direction is gauge-removable while the periodic direction obstructs—the obstruction lives in H^1 of the actual geometry; and the $\mathbb{Z}_2$ -type ladder has solution-space dimension 2 uncoupled (one metric per chain, $e^{\pm 2wi}$) collapsing to 0 at any rung coupling despite zero net winding.

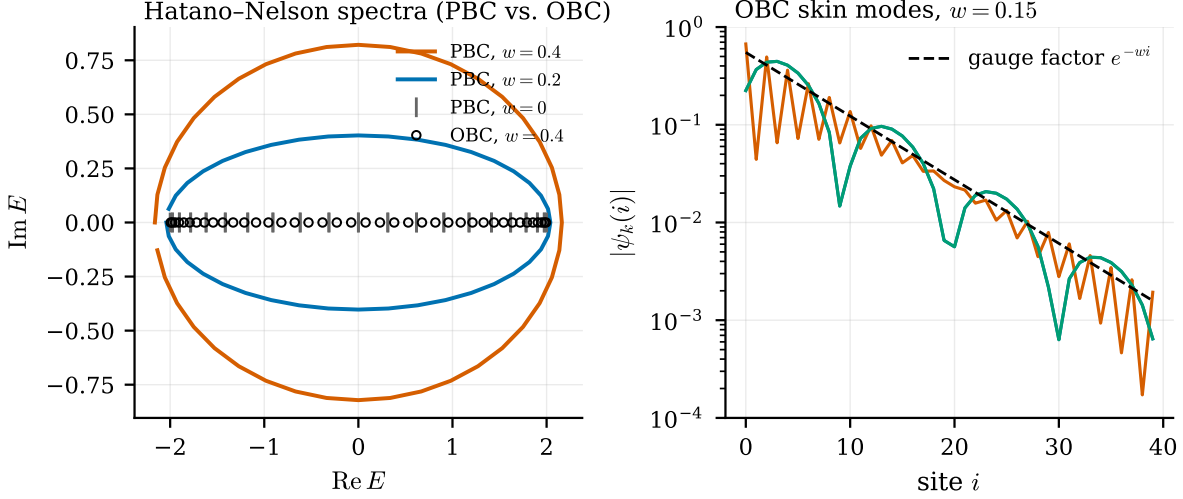


Figure 1: The skin-obstruction theorem (Theorem 3.4). Left: Hatano–Nelson spectra. Nonzero imaginary flux (periodic boundaries, $w \neq 0$) winds the spectrum into a complex loop—no closed process, no fixed beables; the flux-free cases ($w = 0$, or any w with open boundaries) have real spectra. Right: open-boundary skin modes $|\psi_k(i)|$ ride the pure-gauge envelope e^{-wi} (dashed): the skin pileup is precisely the beable-preserving metric $d_i = e^{2wi}$ in disguise, exact at finite N and exponentially degenerate as $N \rightarrow \infty$.

4 The broken phase as an open subsystem

4.1 Broken $\mathcal{PT}$ dynamics is survival-conditioned dynamics

Write (4) as the balanced gain/loss form $H = \cos \theta \mathbf{1} + i \sin \theta \sigma_z + s \sigma_x$. Subtracting the uniform gain,

$$H_{\text{loss}} = H - i \sin \theta \mathbf{1}, \quad e^{-iHt} = e^{\sin \theta t} C(t), \quad C(t) = e^{-iH_{\text{loss}}t}, \quad (6)$$

where $C(t)$ is a contraction (singular values ≤ 1 ; verified to machine precision). The notorious broken-phase runaway is thus a scalar normalization: broken- $\mathcal{PT}$ evolution is the no-leak-conditioned dynamics of a lossy system.

For each t the Halmos dilation [12, 13]

$$U_{\text{big}}(t) = \begin{pmatrix} C(t) & \sqrt{\mathbf{1} - C(t)C(t)^\dagger} \\ \sqrt{\mathbf{1} - C(t)^\dagger C(t)} & -C(t)^\dagger \end{pmatrix} \quad (7)$$

is exactly unitary with $\Gamma_{\text{big}}(0) = \mathbf{1}_4$. Because the correspondence requires only per-time unistochasticity—the family need not compose— $\Gamma_{\text{big}}(t) = |U_{\text{big}}(t)|^2$ is an *ordinary* Barandes indivisible process on four configurations $\{1, 2, a_1, a_2\}$ (two system sites, two “leaked” ancilla configurations), and the reduction

$$\Gamma_{\text{red}}(t) = |C(t)|^2 \quad (8)$$

is a sub-stochastic process (column deficits = leak probabilities), valid at all times in the phase where the closed construction failed totally. (A unitary-embedding perspective on $\mathcal{PT}$ dynamics—a “hidden entangled partner” restoring lost information—was developed by Kawabata, Ashida, and Ueda [30]; the per-time Halmos dilation used here is its process-level counterpart.)

4.2 Exact generator: terminal indivisibility

Theorem 4.1 (terminal indivisibility in the broken phase). *Let $\tilde{\kappa} = \sqrt{\sin^2 \theta - s^2} > 0$ and set (notation local to this theorem and its proof) $c = \cosh \tilde{\kappa} t$, $\sigma = \sinh(\tilde{\kappa} t)/\tilde{\kappa}$,*

$$p = c + \sin \theta \sigma, \quad q = c - \sin \theta \sigma, \quad r = s \sigma.$$

Then

$$\Gamma_{\text{red}}(t) = e^{-2 \sin \theta t} \begin{pmatrix} p^2 & r^2 \\ r^2 & q^2 \end{pmatrix}, \quad \det \Gamma_{\text{red}} = e^{-4 \sin \theta t} (1 - 2r^2), \quad (9)$$

and the generator $L = \dot{\Gamma}_{\text{red}} \Gamma_{\text{red}}^{-1}$ has the exact off-diagonal entries

$$L_{12} = \frac{2spr}{1 - 2r^2}, \quad L_{21} = \frac{2sqr}{1 - 2r^2}. \quad (10)$$

Consequently:

- (i) the process is P -divisible precisely on $[0, t_c)$ with $t_c = \tilde{\kappa}^{-1} \operatorname{arcsinh}(\tilde{\kappa}/(s\sqrt{2}))$, which is exactly the time at which $\det \Gamma_{\text{red}}$ crosses zero;
- (ii) $L_{12} < 0$ strictly for all $t > t_c$: permanent (terminal) indivisibility;
- (iii) $L_{12} \rightarrow -(\sin \theta + \tilde{\kappa})$ as $t \rightarrow \infty$;
- (iv) L_{21} returns positive at $t_0 = \tilde{\kappa}^{-1} \operatorname{artanh}(\tilde{\kappa}/\sin \theta) > t_c$: late-time indivisibility is carried entirely by transitions into the gain site;
- (v) the same formulas continue to the unbroken phase ($\tilde{\kappa} \rightarrow i\kappa$), where $1 - 2r^2 = 1 - 2(s/\kappa)^2 \sin^2 \kappa t$ with $(s/\kappa)^2 > 1$ oscillates through zero forever (recurrent indivisibility windows of period π/κ), and to the exceptional point, where everything is polynomial with $r = st$ and $t_c = 1/(s\sqrt{2})$, continuously in s .

Proof. Since $A = i \sin \theta \sigma_z + s \sigma_x$ obeys $A^2 = (s^2 - \sin^2 \theta) \mathbf{1} = -\tilde{\kappa}^2 \mathbf{1}$, $e^{-iAt} = c \mathbf{1} - i\sigma A$, giving (9) entrywise. With $r' = sc$ and $p' = \tilde{\kappa}^2 \sigma + \sin \theta c$ one finds the Wronskian-type identities

$$r'p - p'r = r'q - q'r = s(c^2 - \tilde{\kappa}^2 \sigma^2) = s,$$

constants in t ; inserting them into $L = \dot{\Gamma} \tilde{\Gamma}^{-1} - 2 \sin \theta \mathbf{1}$ (the scalar factor shifts only the diagonal) yields (10). Statements (i)–(v) follow by inspection of signs: $p, r > 0$ always, $1 - 2r^2$ crosses zero once (hyperbolic) at t_c , q crosses zero once at $t_0 > t_c$, and $L_{12} \rightarrow -(\sin \theta + \tilde{\kappa})$ from the asymptotics $c \sim \tilde{\kappa} \sigma \sim \frac{1}{2} e^{\tilde{\kappa} t}$. $\square$

The predictions are exact against brute-force numerics (Fig. 2; closed forms to 10^{-14} , generator to 10^{-11} ; the observed floor -1.207 at $s = 0.5$ is $-(\sin \frac{\pi}{4} + 0.5) = -1.2071$; the observed first indivisible time 1.35 on a 0.05 grid is $t_c = 1.3170$).

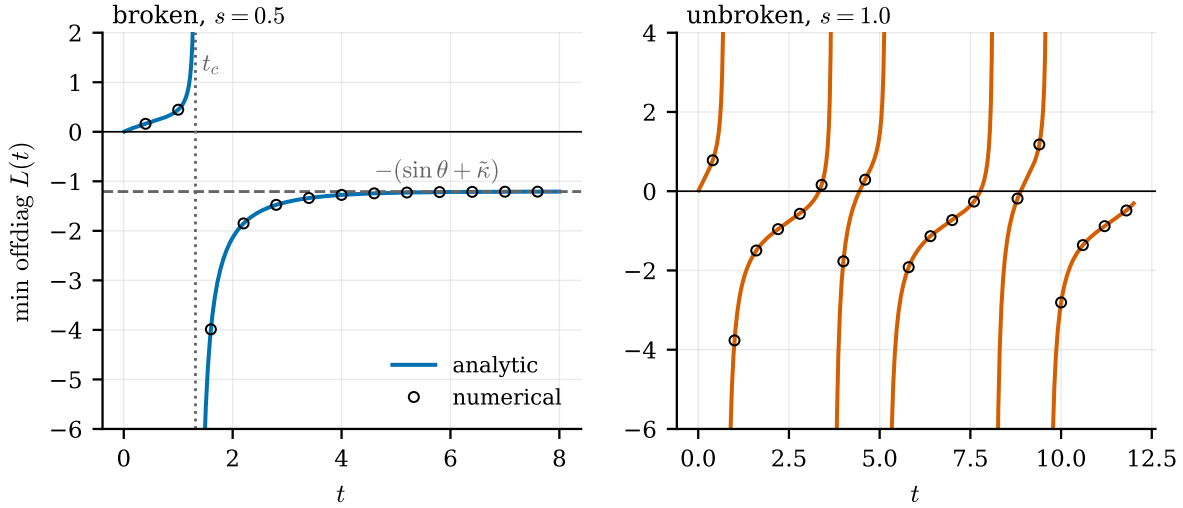


Figure 2: Minimum off-diagonal entry of the generator $L(t) = \dot{\Gamma}_{\text{red}} \Gamma_{\text{red}}^{-1}$ of the reduced process (8): analytic closed forms (10) (curves) versus brute-force numerics (circles). Left: broken phase ($s = 0.5$)—divisibility is lost at t_c , exactly where $\det \Gamma_{\text{red}}$ crosses zero, and never recovers; the generator approaches the strictly negative floor $-(\sin \theta + \tilde{\kappa})$ (Theorem 4.1): *terminal* indivisibility. Right: unbroken phase ($s = 1.0$)—the same closed forms continued to trigonometric functions give *recurrent* windows of divisibility and indivisibility at the Rabi period (curves interrupted where $\det \Gamma_{\text{red}}$ passes through zero).

4.3 The two non-Markovianities dissociate

For open systems, non-Markovianity is commonly diagnosed by information backflow (increase of trace/total-variation distance between two evolutions) [14, 15, 16]. On the classical marginal with the leak recorded as a third outcome, the total-variation distance is monotone in *both* phases (no classical backflow at all); on the survival-conditioned (post-selected) process, the late-window backflow vanishes identically for every broken s and jumps at the EP (e.g. $0.000 \rightarrow 1.32$ across $s_* = 0.7071$ at $s = 0.72$, window $t \in [20, 40]$)—the stochastic-level analogue of the reversible–irreversible criticality of information flow established for $\mathcal{PT}$ systems by Kawabata, Ashida, and Ueda [30]. Meanwhile, Barandes indivisibility is present in *both* phases; what changes at the EP is its temporal structure, recurrent $\leftrightarrow$ terminal (the terminal-indivisibility theorem, Theorem 4.1). Figure 3 contrasts the two diagnostics across the transition: the backflow order parameter switches at the EP, while the indivisibility onset $t_c(s)$ is finite and *continuous* through it.

Thus the broken phase is *backflow-Markovian in the late window yet permanently Barandes-indivisible*: the two notions genuinely dissociate, indivisibility being strictly finer. That the relation between divisibility- and information-flow-based notions of Markovianity is subtle for quantum channels—equivalent for invertible maps [31], genuinely inequivalent for noninvertible ones [32]—is known; the $\mathcal{PT}$ dimer makes the dissociation exact, phase-locked, and—in the broken phase—permanent, at the level of the classical configuration process. A closed Hermitian system, being quasi-periodic, can only be recurrently indivisible; terminal indivisibility with monotone leak is the stochastic signature of the broken phase. (At $n = 2$ every lossy two-site

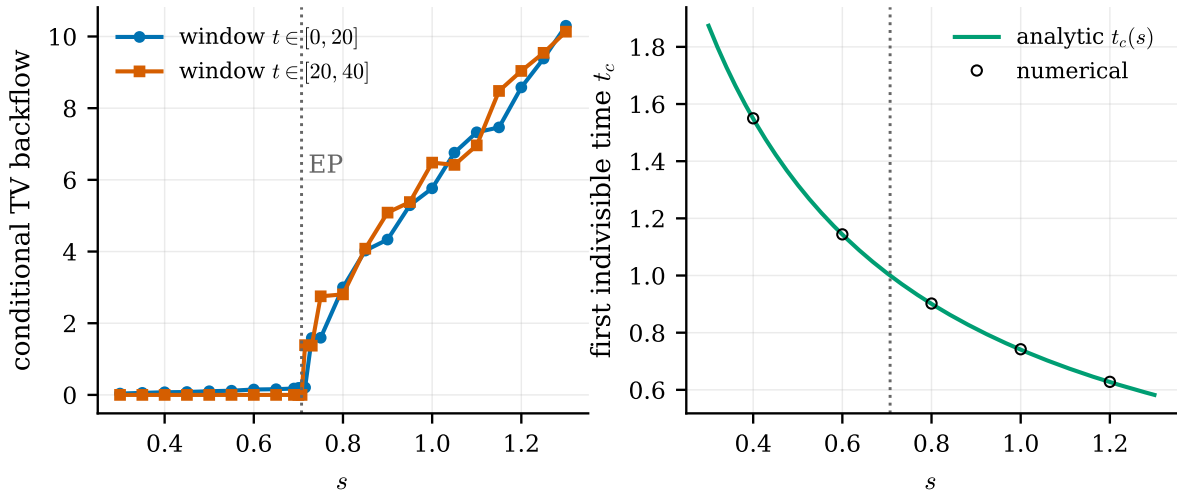


Figure 3: The two non-Markovianities across the $\mathcal{PT}$ transition (EP at $s_* = \sin \frac{\pi}{4} \approx 0.707$, dotted line). Left: conditional total-variation backflow in an early ($t \in [0, 20]$) and a late ($t \in [20, 40]$) window; the late-window backflow vanishes identically in the broken phase and switches on at the EP (the apparent smoothness just above s_* is the diverging Rabi period π/κ meeting the finite window). Right: the first indivisible time $t_c(s)$ of Theorem 4.1, analytic in both phases and continuous through the EP (circles: numerical onset). Indivisibility is present on both sides; backflow is not.

Hamiltonian *with degenerate site frequencies* is a shifted $\mathcal{PT}$ dimer, so there “overdamped open dynamics” and “broken $\mathcal{PT}$ ” coincide; a detuned pair adds a real σ_z component the $\mathcal{PT}$ form does not carry.)

5 The exceptional point is a coordinate singularity

At $s_* = \sin \theta$ the dimer is defective (a Jordan block; numerically $\text{rank}(H - \cos \theta \mathbf{1}) = 1$), the naive process grows polynomially ($|U|^2$ column sums $\sim t^2$; fitted exponent 1.95), and the two-dimensional space of Hermitian intertwiners consists of one *singular positive-semidefinite rank-one ray* and one indefinite direction: the positive-definite cone is exited exactly through its boundary, and ρ does not exist.

Approaching from the unbroken side, $s = s_*(1 + \varepsilon)$, $\kappa \simeq \sin \theta \sqrt{2\varepsilon}$, we find clean power laws (Fig. 4; log-log fits over $\varepsilon = 10^{-1} \dots 10^{-7}$, all within ± 0.0025):

quantity	scaling	fitted slope
$\text{cond}(\eta)$	$\varepsilon^{-1} \sim \kappa^{-2}$ (Petermann [20])	−0.998
$\text{cond}(\rho)$ (beable-map anisotropy)	$\varepsilon^{-1/2} \sim \kappa^{-1}$	−0.499
$\ \Gamma_\varepsilon(t) - \mathbf{1}\ $ at fixed t	$\varepsilon \sim (\kappa t)^2$	+0.999

At fixed laboratory time the metric-corrected process therefore *freezes*: $h \rightarrow \cos \theta \mathbf{1}$ and $\Gamma_\varepsilon(t) \rightarrow \mathbf{1}$. A regularized lab-time limit exists but is trivial—and is not the EP dynamics, which grows like t^2 . On the critically rescaled clock $\tau = \kappa t$ the process converges (to 10^{-10} between

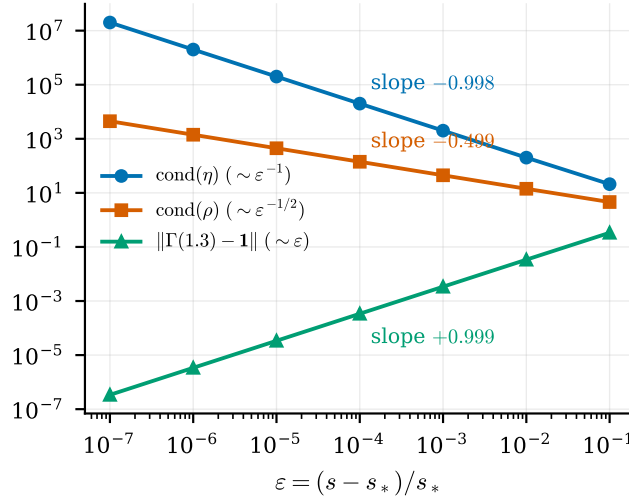


Figure 4: Collapse of the closed-system (metric) description approaching the exceptional point from the unbroken side, $s = s_*(1 + \varepsilon)$. The metric condition number diverges with the Petermann scaling $\varepsilon^{-1} \sim \kappa^{-2}$, the beable-map anisotropy as $\varepsilon^{-1/2} \sim \kappa^{-1}$, while the metric-corrected process at fixed laboratory time *freezes*, $\|\Gamma_\varepsilon(t) - \mathbf{1}\| \sim \varepsilon \sim (\kappa t)^2$.

$\varepsilon = 10^{-4}$ and 10^{-6}) to the universal indivisible Rabi process

$$\Gamma_{\text{lim}}(\tau) = \begin{pmatrix} \cos^2 \tau & \sin^2 \tau \\ \sin^2 \tau & \cos^2 \tau \end{pmatrix}, \quad (11)$$

with Rabi axis exactly σ_x ; but the beable map identifying its states with the original configurations diverges ($\text{cond } \rho \sim \kappa^{-1}$): the survivor is abstract.

By contrast, every open-system quantity of Sec. 4 (reduced-process entries, survival, indivisibility generator floor) varies smoothly and monotonically through s_* . *The EP is a singularity of the similarity-to-Hermitian dictionary, not of the stochastic process*: existence of an indivisible process is governed by the dilation, and the deflation of Proposition 2.1 fails at the EP only because its coordinates do.

6 The Pais–Uhlenbeck oscillator

The PU Lagrangian $L = \frac{\gamma}{2} [\ddot{x}^2 - (\omega_1^2 + \omega_2^2) \dot{x}^2 + \omega_1^2 \omega_2^2 x^2]$ has the Ostrogradski Hamiltonian (with $q_1 = x$, $q_2 = \dot{x}$)

$$H_{\text{PU}} = p_1 q_2 + \frac{p_2^2}{2\gamma} + \frac{\gamma}{2} (\omega_1^2 + \omega_2^2) q_2^2 - \frac{\gamma}{2} \omega_1^2 \omega_2^2 q_1^2, \quad (12)$$

which is *Hermitian* on the real configuration space $(x, \dot{x})$, with the ghost spectrum $E = \omega_1(n_1 + \frac{1}{2}) - \omega_2(n_2 + \frac{1}{2})$: real, unbounded below [9, 8].

6.1 The ghost realization is a valid fixed-beable process

On a discretized $(x, \dot{x})$ grid ($44^2 = 1936$ configurations, spectral derivatives, $\omega_1 = 1$, $\omega_2 = 0.5$, $\gamma = 1$) the operator (12) is exactly Hermitian; $\Gamma = |U|^2$ is doubly stochastic to 2×10^{-14} and

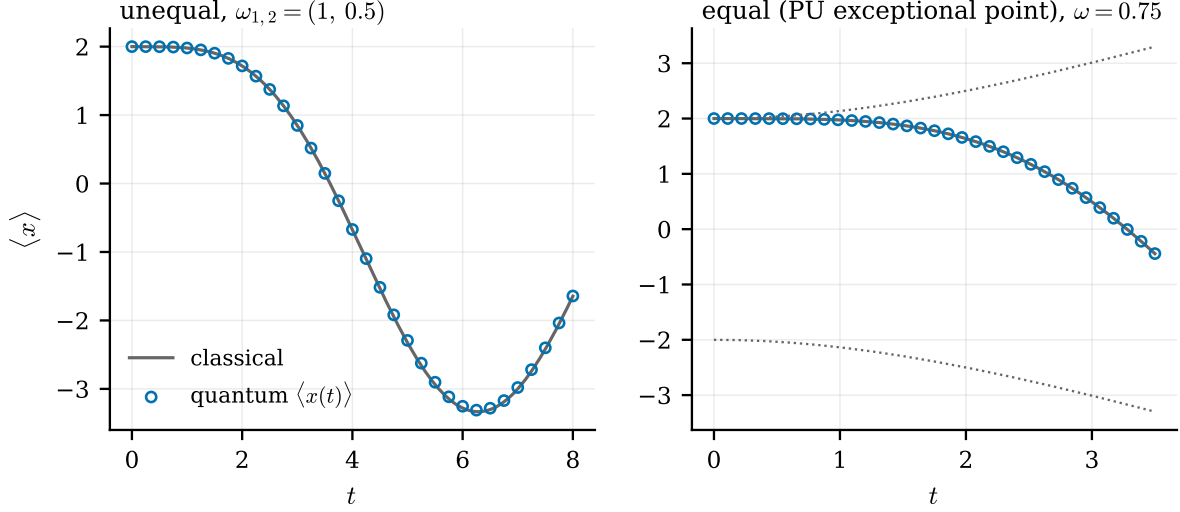


Figure 5: The ghost realization of the free Pais–Uhlenbeck oscillator as a stochastic process on physical configurations: $\langle x(t) \rangle$ of the grid process (circles) versus the exact classical fourth-order trajectory (line). Left: unequal frequencies. Right: equal frequencies—the PU exceptional point—where the classical normal-mode matrix is defective and the solution grows secularly, $x(t) = 2 \cos \omega t + \omega t \sin \omega t$ (dotted: envelope $\pm \sqrt{4 + (\omega t)^2}$); the Hermitian realization crosses the point smoothly while Γ remains exactly doubly stochastic.

genuinely indivisible ($\|\Gamma(2t) - \Gamma(t)^2\| \approx 2.7$); Hermiticity makes (K1)–(K4) hold trivially with $\eta = \mathbf{1}$, so the beables are fixed. The realization is validated dynamically: Ehrenfest is exact for quadratic Hamiltonians, and $\langle x(t) \rangle$ tracks the exact classical fourth-order trajectory to 3×10^{-3} over half the evolution window (residual from packet tails at the box walls; Fig. 5).

The spectrum of the grid operator runs from -26 to $+140$ and is *completely irrelevant to the process*: at the free level the Ostrogradski ghost is an energy-accounting pathology, invisible to the stochastic representation.

At equal frequencies (the PU exceptional point) the classical normal-mode matrix is defective (eigenvector condition number $\sim 10^8$; secular $t \sin \omega t$ growth), yet the Hermitian realization passes through smoothly: $\langle x(t) \rangle$ tracks $2 \cos \omega t + \omega t \sin \omega t$ to 9×10^{-4} and Γ stays exactly doubly stochastic—the PU-scale confirmation of Sec. 5.

6.2 The no-ghost quantization exits the sample space

Mannheim’s positive spectrum requires re-quantizing the ghost mode with a vacuum satisfying $b^\dagger \Omega = 0$, i.e. $\Omega \propto e^{+y^2/2}$: normalizable only on a rotated contour [8]. This is a *domain change*, not a similarity (a similarity cannot flip the spectrum), and it has no shadow at any truncation: the null vector of the truncated b is $|0\rangle$ at every N (the true vacuum, truncation-stable), while the null vector of the truncated $b^\dagger$ is $|N-1\rangle$, pinned to the truncation edge and escaping to infinite energy; restricted to any fixed low-energy sector, $\sigma_{\min}(b^\dagger) = 1$ exactly. The contour-rotation operator $R = e^{(\pi/2)D}$, $D = \frac{1}{2}(yp + py) = \frac{i}{2}(a^{\dagger 2} - a^2)$, has $\log_{10} \text{cond}(R_N)$ growing at ≈ 1.2 decades per Fock level (53 decades at $N = 48$), so the would-be metric $\sim (RR^\dagger)^{-1}$ is unbounded with unbounded inverse: the Sec. 5 collapse at every scale at once, with no critical-rescaling

survivor.

Verdict. The stochastic–quantum correspondence and the $\mathcal{PT}$ no-ghost construction resolve the free ghost in incompatible currencies: Barandes keeps real beables and pays in negative energies; Mannheim buys positive energies and pays with the sample space itself. (A related construction by Salvio and Strumia [33] obtains positive energies and normalizable wave functions at the price of a negative-norm configuration-space inner product—again incompatible with a probability measure over configurations.) The dynamics-dependent beable rotation of Sec. 2 escalates, in the PU limit, to beable *non-existence*.

7 Interacting ghosts: conservative, transient, explosive

Does the ghost instability, once interactions are switched on, become a *stochastic* pathology? The answer arrives in stages.

7.1 Quadratic interactions: a valid process turns transient

Consider the Hermitian cascade Hamiltonian

$$H = \omega_1 a^\dagger a - \omega_2 b^\dagger b + g(a^\dagger b^\dagger + ab), \quad (13)$$

whose normal twin has $+\omega_2 b^\dagger b$. The Heisenberg mode matrix has eigenvalues $\bar{\omega} \pm \sqrt{\delta^2/4 - g^2}$ (ghost; $\delta = \omega_1 - \omega_2$, $\bar{\omega} = (\omega_1 + \omega_2)/2$), so the ghost sign collapses the instability threshold from the *sum* frequency, $g > (\omega_1 + \omega_2)/2$ (normal), to the *difference* frequency, $g > |\omega_1 - \omega_2|/2$: zero at resonance. At resonance the vacuum cascades with the exact two-mode-squeezing laws

$$\langle n_1(t) \rangle = \langle n_2(t) \rangle = \sinh^2(gt), \quad P(n_1 \leq K) = 1 - \tanh^{2K+2}(gt) \rightarrow 0, \quad (14)$$

both matched numerically to ~ 4 decimals in the resolved regime ($\langle n \rangle \lesssim$ a few, before the 34^2 Fock truncation sets in at the last tabulated time) (Fig. 6; Fock truncation 34^2 , $g = 0.15$; detuned growth consistent with $(g^2/\kappa^2) \sinh^2 \kappa t$, $\kappa = \sqrt{g^2 - \delta^2/4}$), while Γ remains doubly stochastic to 2×10^{-14} and the energy books stay exactly balanced ($E_{\text{normal}} = +8.301$, $E_{\text{ghost}} = -8.301$, $\langle H \rangle = 0$ conserved).

Vacuum decay is a configuration-space cascade with balanced energy books: the process is valid and indivisible but *transient*—it permanently escapes every finite region at rate $2g$ —which is the first stage at which the ghost becomes visible in the process’s own currency.

7.2 Malicious nonlinearities: finite-time escape

Classically, the ghost pair $\frac{1}{2}(p_1^2 + \omega_1^2 x_1^2) - \frac{1}{2}(p_2^2 + \omega_2^2 x_2^2) + \lambda x_1^2 x_2^2$ is mostly bounded at moderate amplitude (benign islands [11]), while the quartic PU,

$$\ddot{x} + (\omega_1^2 + \omega_2^2)\ddot{x} + \omega_1^2 \omega_2^2 x = \frac{4\lambda}{\gamma} x^3, \quad (15)$$

exhibits robust *finite-time* blowup for both signs of λ at every amplitude tested ($t_* = 3.3\text{--}6.9$); for $\lambda > 0$ the dominant balance is $x \sim A(t_* - t)^{-2}$, $A = \sqrt{30\gamma/\lambda}$ (the $\lambda < 0$ escape proceeds by a different mechanism, taken up in Sec. 9). Quantum-mechanically, a lattice can transport the accelerating wavefront only while the required classical speed stays below the Nyquist momentum $\pi/\Delta x$; box-size drifts of the leak onset correlate exactly with $v_{\text{cl}}(\text{edge})/k_{\text{max}} = 1.4, 2.4, 3.4$ and

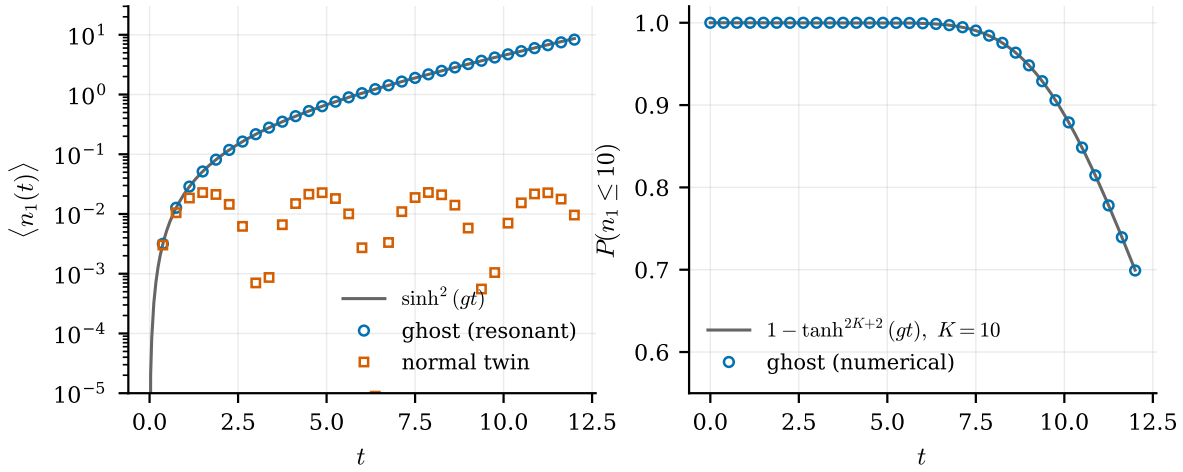


Figure 6: The quadratic ghost cascade (13) at resonance ($\omega_1 = \omega_2$, $g = 0.15$): a valid, exactly doubly stochastic process that is *transient*. Left: occupation growth $\langle n_1(t) \rangle = \sinh^2(gt)$ (line: exact law; circles: Fock-space numerics), while the normal-sign twin (squares) stays bounded below 3×10^{-2} —the instability threshold collapses from the sum to the difference frequency. Right: escape from the occupation ball $n_1 \leq 10$ follows the exact two-mode-squeezing law (14).

are lattice confinement, not physics (a fact that itself echoes Sec. 8: a non-essentially self-adjoint operator has no faithful finite truncation). In the resolved regime, quantum probability arrives at every probed radius at times bounded by the classical finite-time law, with bulk weight, while the free control never passes $r \approx 7$.

In the standard vocabulary of stochastic processes, the hierarchy is: free/normal-sign = *conservative*; ghost with quadratic (or benign nonlinear) interaction = *transient*; ghost with malicious nonlinearity = *explosive* (mass leaves the state space in finite time). The Ostrogradski ghost is a recurrence-classification problem. Complementary operator-theoretic and dynamical stability results for interacting ghosts—the global-stability classification program of Errasti Díez, Gaset Rifà, and Staudt [34] and its multidimensional classical-scattering extension by Held [35], conserved quantities with discrete spectrum [36], and exact moment bounds [37]—identify interaction structures in which the cascade is arrested; in the present language, those are mechanisms enforcing recurrence (none of these works uses the stochastic/recurrence currency itself).

8 The explosion dictionary

The mechanism of malicious escape is one-dimensional: motion along a direction where the effective potential falls super-quadratically. For this mechanism the correspondence between the three descriptions can be made a theorem.

Theorem 8.1 (explosion dictionary, 1D mechanism model). *Let $H = \frac{1}{2}p^2 + V$ on the line with an escaping end ($V \rightarrow -\infty$ eventually monotonically and C^1 with $|V'|/|V|^{-3/2}$ bounded, the standard regularity under which the limit-point/limit-circle criteria below apply—the mechanism model; Rauch–Reed’s oscillatory one-dimensional counterexamples [26] show this hypothesis cannot be*

dropped), and define the classical time of flight $T(E) = \int^\infty dx / \sqrt{2(E - V(x))}$.

- (1) If $T = \infty$ (e.g. $V \gtrsim -Kx^2$), the end is limit point, and—with the non-escaping end confining, hence also limit point— H is essentially self-adjoint [22], there is a unique unitary $U(t)$ and a unique conservative process.
- (2) If $T < \infty$ (e.g. $V \lesssim -Kx^{2+\epsilon}$), the end is limit circle [22], by the (WKB-asymptotic) identity

$$|\psi_\pm(x)|^2 \approx [2(E - V(x))]^{-1/2} \implies \int^\infty |\psi_\pm|^2 dx \approx T : \quad (16)$$

the deficiency solution’s norm is, to WKB accuracy, the classical time of flight (*the limit-point/limit-circle dichotomy itself is exact; the identity quantifies it semiclassically*). Consequently:

- (a) *there is no canonical unitary dynamics: self-adjoint extensions form a $U(1)$ family per escaping end ($U(2)$ for two ends, including completions that transmit through infinity), i.e. boundary data at infinity [22];*
- (b) *every extension is conservative but returns probability from infinity after the finite classical round trip; its spectrum is discrete with local spacing, semiclassically, $\Delta E \simeq \pi/T_{\text{cross}}(E)$, T_{cross} the classical traversal time;*
- (c) *the regulator-independent object is the minimal (Feller) semigroup [25]: a sub-stochastic, explosive process losing mass at the classical escape rate;*
- (d) *one integral decides classical completeness, quantum essential self-adjointness (Weyl [21]), and stochastic conservativity (Feller exit boundary).*

The rigorous inputs are Weyl’s alternative and the limit-point/limit-circle criteria of [22] (Thms. X.8–X.10); the WKB norms and the semiclassical spacing law are used at their standard validity.

Numerical verification (representative $V = -x^4$, marginal control $V = -x^2$): the spacing law matches to 3.5% or better in all eight (V, L) cases tested and to 0.3% at the largest box for $V = -x^4$ (Table 1); the eigenvalue nearest $E = 10$ (a tracked-level probe, distinct from the spacing window at $E = 11$) sweeps erratically with the wall position (no completion has an $L \rightarrow \infty$ limit) while the spacing is stable; a 1D absorbing-boundary evolution (where the lattice barrier is beatable) absorbs 98.6–98.8% of the probability by $t = 1$ *independently of box size*, with onset saturating toward the classical arrival ($T_{\text{esc}} = 0.4635$; Fig. 7, left); and two completions of the same H —Dirichlet wall versus periodic identification—both revive $\approx 95\%$ of the probability after the finite round trip but from *opposite sides* (reflection versus a $+\infty \rightarrow -\infty$ “wormhole”; Fig. 7, right), an observable difference encoded nowhere in H .

9 Multidimensional theory: channels and the S -matrix at infinity

The naive dictionary “classical incompleteness $\Leftrightarrow$ quantum non-self-adjointness” fails in general beyond the monotone one-dimensional mechanism model: Rauch and Reed constructed one-dimensional counterexamples in both directions [26], and in $n \geq 2$ the correct structure is channel-wise. Throughout, H is defined on C_0^∞ and deficiency indices refer to its closure.

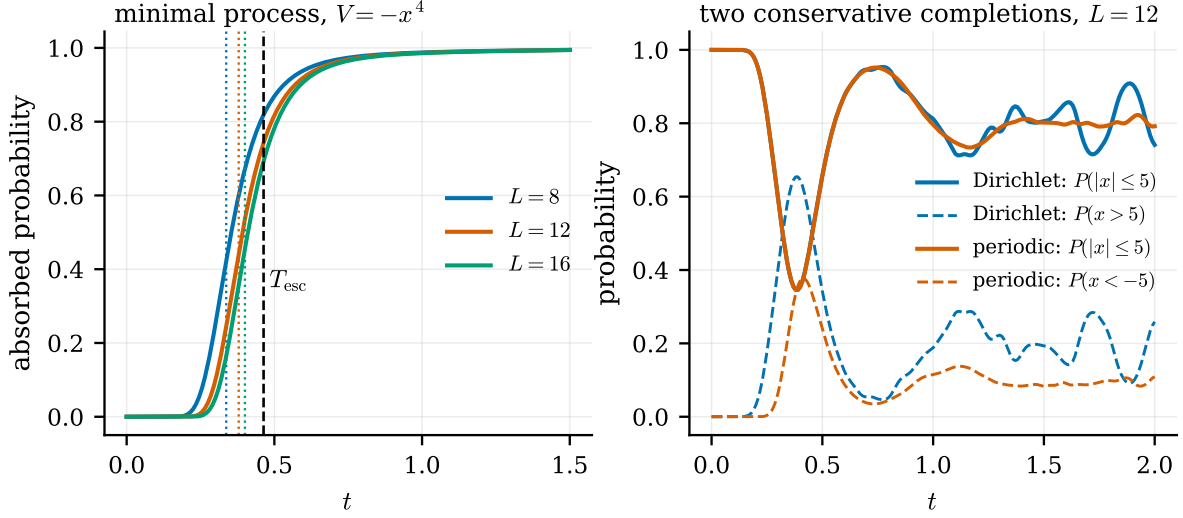


Figure 7: Explosion and its completions for $V = -x^4$ (the explosion dictionary, Theorem 8.1). Left: the minimal (absorbed) process is box-independent—the absorbed probability for $L = 8, 12, 16$ collapses onto one curve with onset tracking the classical arrival at each absorber edge (dotted verticals) and saturating at the finite escape time T_{esc} (dashed): probability reaches infinity at finite time. Right: two conservative completions of the same symmetric operator. Both revive $\approx 95\%$ of the probability after the finite round trip “through infinity,” but the Dirichlet wall returns it from the right (reflection) while the periodic identification returns it from the left (transmission, $+\infty \rightarrow -\infty$): the completion is boundary data at infinity, not physics contained in H .

Theorem 9.1 (quadratic ghosts are safe). *Every at-most-quadratic Hamiltonian on $\mathbb{R}^n$ —including the indefinite cascade (13) at any coupling, even in the unstable phase, and the free PU operator (12)—is essentially self-adjoint on the span of Hermite functions.*

Proof. A quadratic H maps Hermite level N into levels $\leq N + 2$ with coefficients $O(N)$, so $\|H^n \psi_N\| \leq (2C)^n (N/2 + n)! / (N/2)!$ and $\sum_n t^n \|H^n \psi_N\| / n!$ converges for small t : every Hermite function is an analytic vector. Nelson’s analytic vector theorem [24] applies. $\square$

Thus instability never breaks uniqueness or conservativity: the transient/explosive dichotomy of Sec. 7 is rigorous on the transient side.

Theorem 9.2 (channel additivity of deficiency). *Let $H = h_1 \otimes \mathbf{1} + \mathbf{1} \otimes h_\perp$ on $L^2(\mathbb{R} \times \mathbb{R}^m)$, with $h_1 = \frac{1}{2}p^2 + V_1$ super-quadratically falling at every end (limit circle at both $\pm\infty$) and $h_\perp$ self-adjoint with orthonormal eigenbasis $\{\varphi_n, E_n\}$. Then for every n and every solution u of $(h_1 - (\pm i - E_n))u = 0$ (all of which are L^2 , since the limit-circle property at each end is independent of the spectral parameter), $u \otimes \varphi_n \in \ker(H^* \mp i)$. Hence the deficiency indices of H are (∞, ∞) .*

Proof. H^* acts distributionally, and $H(u \otimes \varphi_n) = ((\pm i - E_n)u) \otimes \varphi_n + E_n u \otimes \varphi_n = \pm i u \otimes \varphi_n \in L^2$, so $u \otimes \varphi_n \in D(H^*)$ and is a deficiency vector; vectors with different n are orthogonal. $\square$

Corollary 9.3. $-\frac{1}{2}\Delta - c|x|^\alpha$ on $\mathbb{R}^d$ with $\alpha > 2$ has deficiency indices (∞, ∞) (partial-wave separation; the centrifugal terms are subquadratic).

L	$V = -x^4$ (malicious)		$V = -x^2$ (marginal)	
	measured ΔE	π/T_{cross}	measured ΔE	π/T_{cross}
6	2.559	2.609	1.619	1.640
9	2.365	2.449	1.290	1.289
12	2.345	2.377	1.106	1.112
16	2.318	2.325	0.964	0.975

Table 1: Level spacing of the wall-regularized completions near $E = 11$ versus the classical law $\pi/T_{\text{cross}}(E, L)$. For the malicious potential the traversal time to infinity is finite, so the spacing saturates: every conservative completion reflects off infinity at a finite recurrence time. For the marginal control the spacing decays toward the continuum.

Theorem 9.4 (non-separable channel mixing). *Let $H_0 = \frac{1}{2}p_2^2 - |x_2|^\alpha + h_\perp(x_1)$, $\alpha > 2$, and $B = \mu f(x_1)g(x_2)$ with f, g bounded and real-valued. Then $H_0 + B$ has the same (infinite) deficiency indices as H_0 , while B genuinely mixes the transverse channels.*

Proof. Bounded symmetric perturbations preserve deficiency indices [23]. □

Physical content. The conservative completions of a malicious multidimensional ghost are parametrized by unitaries between infinite-dimensional deficiency spaces: a genuine *S-matrix at infinity*, which may scramble the transverse state upon “reflection from infinity”. Numerically (four-channel model, $V_1 = -|x|^{5/2}$, bounded mixing): the wall-regularized level count in a window matches the channel sum $\sum_n \int T_{\text{cross}}(E - E_n, L) dE/\pi$ (34 vs 32.3 at $L = 8$; 37 vs 37.1 at $L = 12$), at $\mu = 0$ the spectrum is exactly a union of shifted 1D ladders (deviation 10^{-12}), switching on the bounded coupling leaves the counts essentially unchanged ($34 \rightarrow 34$ at $L = 8$, $37 \rightarrow 36$ at $L = 12$, a one-level window-edge shift within count granularity; Kato stability made visible)—and a packet sent to infinity in the transverse ground state returns as a channel mixture *whose content depends on where infinity was regularized* (Fig. 8): fractions (0.583, 0.346, 0.058, 0.014) for $L = 6$ versus (0.401, 0.421, 0.143, 0.035) for $L = 9$, with 96%/93% of the probability returning at the classical round-trip times.

Remark 9.5 (an obstruction specific to ghosts). For indefinite $H_0 = h_1 \otimes \mathbf{1} + \mathbf{1} \otimes h_\perp$ with h_1 unbounded below, transverse observables are *not* H_0 -bounded: joint spectral points (λ, μ) with $\lambda \approx -\mu$ make $\|H_0\psi\|$ small while $\|h_\perp\psi\|$ is large. The standard Kato–Rellich transfer therefore fails even for mildly unbounded couplings—a second, ghost-specific obstruction beyond non-separability.

Open Problem 9.6. Determine the deficiency indices of the exact quartic Pais–Uhlenbeck operator, $H_{\text{PU}} + \lambda q_1^4$ of (12), and of the ghost pair with coupling $\lambda x_1^2 x_2^2$, on $C_0^\infty(\mathbb{R}^2)$. Both escape the hypotheses above (unbounded diagonal coupling; degenerate/indefinite kinetic form). Either outcome is significant: infinite deficiency would complete the explosion dictionary for the physical operators, while essential self-adjointness would be a “quantum completion” in the sense of [26]—interactions curing the ghost’s explosion.

Numerical evidence. A regulator-dependence campaign (Fock-space compressions, which regulate position and momentum symmetrically—the PU escape accelerates without bound—with the matched-eigenvalue-drift diagnostic calibrated on limit-circle, continuous-spectrum, and discrete essentially-self-adjoint controls, and on the free PU operator as the two-dimensional

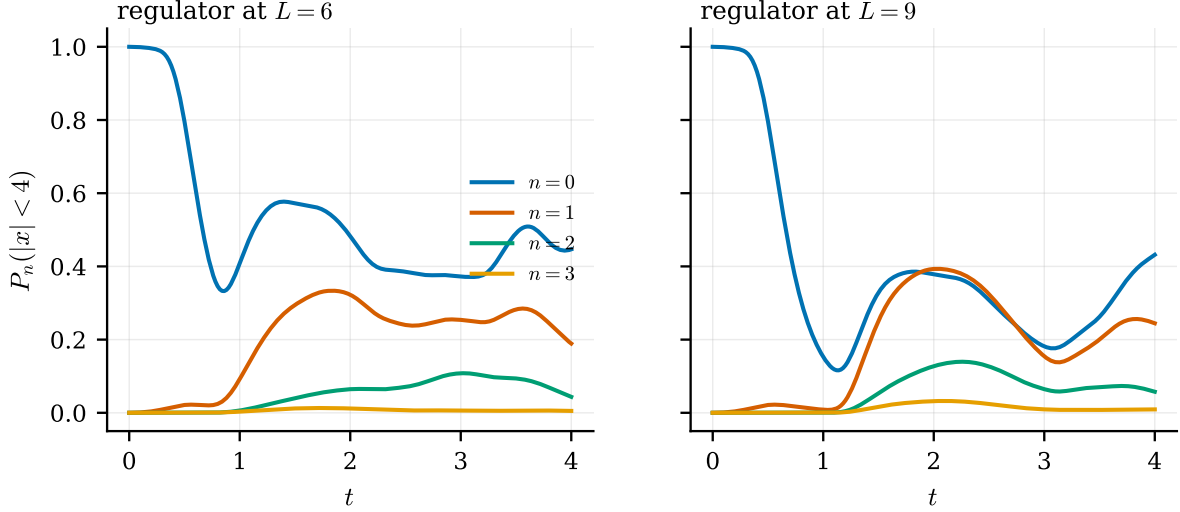


Figure 8: The S -matrix at infinity made visible (four-channel malicious model of the bounded-mixing theorem, Theorem 9.4). Channel-resolved probability in the interior region $|x| < 4$ for a packet launched toward infinity in the transverse ground state $n = 0$. The packet reflects off the regulator and returns as a channel *mixture*; comparing the two regulator positions ($L = 6$, left; $L = 9$, right), the returned mixture differs substantially. In the $L \rightarrow \infty$ limit the reflection law is a genuine unitary between infinite-dimensional deficiency spaces (the channel-additivity theorem, Theorem 9.2): conservative completions scramble the transverse state with data injected at infinity.

baseline) points to a split verdict. The quartic PU shows persistent $O(1)$ drift that does not decay with truncation, at two different basis scales—the limit-circle signature, at $\sim 5\times$ the free-PU baseline: evidence *against* essential self-adjointness, i.e. for the explosion dictionary applying to the physical operator. The ghost pair with $\lambda > 0$ sits at the essentially-self-adjoint baseline, as predicted by its Born–Oppenheimer channels $W_n(x_2) = \frac{1}{2}\omega_2^2 x_2^2 - (n + \frac{1}{2})\sqrt{\omega_1^2 + 2\lambda x_2^2}$, all of which are confining. Unexpectedly, so does the classically malicious $\lambda < 0$ ghost pair—tentative evidence of a genuine Rauch–Reed-type quantum completion, in which zero-point energy walls off the narrow classical escape valley. Evidence, not proof; but it sharpens the problem: prove deficiency for the quartic PU by quasi-mode construction along its escape channel, and test the quantum-completion hypothesis for the ghost pair with valley-adapted bases. (Deficiency indices of related polynomial-oscillator operators—higher-order squeezing—have recently been computed rigorously, including cases where added terms *restore* essential self-adjointness [39]: an operator-theoretic precedent for completion by interaction.)

The drift verdict for the quartic PU can, in fact, be given structural support, together with a falsifiable sign prediction:

Proposition 9.7 (displaced-oscillator identity (exact) and channelization (to $O(k^2)$) of the quartic PU). *In the Fourier-rotated representation ($p_1 \rightarrow y$, $q_1 \rightarrow -p_y$), the displaced-oscillator unitary $U = \exp[-i y p_2 / (\gamma \Omega)]$, $\Omega = \omega_1^2 + \omega_2^2$, gives exactly*

$$U(H_{\text{PU}} + \lambda q_1^4)U^\dagger = T\left(p_y + \frac{p_2}{\gamma\Omega}\right) + \frac{p_2^2}{2\gamma} + \frac{\gamma\Omega}{2}q_2^2 - \frac{y^2}{2\gamma\Omega}, \quad T(p) = \lambda p^4 - \frac{\gamma\Omega_0^2}{2}p^2, \quad (17)$$

with $\Omega_0^2 = (\omega_1\omega_2)^2$ (verified in Fock space via the terminating Baker–Campbell–Hausdorff series to 10^{-9}). The mode-2-diagonal (channel) part is, in the $k = p_y$ representation and up to $O(k^2)$ corrections, minus the Schrödinger operator $\hat{p}^2/(2\gamma\Omega) - \lambda k^4 + (\gamma\Omega_0^2/2)k^2$; interchannel couplings are polynomial of degree ≤ 3 in k , subordinate to the quartic diagonal. Hence for $\lambda > 0$ every channel is limit circle (finite channel time of flight: 1.74 for the parameters used here), and channel additivity as in Theorem 9.2 argues for deficiency indices (∞, ∞) —with the stability question settled at every finite channel number by the finite-channel stability theorem (Theorem 9.8) below. For $\lambda < 0$ every channel is (minus) a confining operator: quantum completion predicted, despite classical finite-time blowup at both signs.

The sign prediction is confirmed by the drift diagnostic: the classically explosive $\lambda < 0$ quartic PU sits near the essentially-self-adjoint baseline (ratios 0.17–0.46 at two basis scales, none leaving the essentially-self-adjoint baseline band), separated from the $\lambda > 0$ limit-circle signature (the extended truncations give $O(1)$ ratios—3.77 at $N = 48 \rightarrow 56$, 1.06 at $N = 56 \rightarrow 64$ —several times to an order of magnitude above the ~ 0.2 e.s.a. baseline; a single early window at one basis scale dips to 0.34), while the $\lambda < 0$ ghost pair gives 0.49 at $N = 48 \rightarrow 56$ and 0.32 at $N = 56 \rightarrow 64$ —the first value above the control bands but both well below the limit-circle signature. The $\lambda < 0$ quartic PU thus joins the $\lambda < 0$ ghost pair as a Rauch–Reed-type quantum-completion candidate: in both cases the classical escape appears to be walled off quantum mechanically, whereas the $\lambda > 0$ quartic PU escapes through momentum space along limit-circle channels.

The stability of channel deficiency under the interchannel couplings—the gap flagged after Proposition 9.7—can in fact be closed at every finite channel number, because the couplings are not generic subordinate couplings: they are functions of the single operator p_2 .

Theorem 9.8 (finite-channel deficiency stability). *For every M , the M -channel Galerkin model of the channelized quartic PU ($\lambda > 0$),*

$$S_M = \frac{\hat{p}^2}{2\gamma\Omega} - T(k\mathbf{1} + cP_{2,M}) - \sqrt{\Omega}\left(N + \frac{1}{2}\right)\Big|_M, \quad c = \frac{1}{\gamma\Omega},$$

with $P_{2,M}$ the truncated momentum of mode 2, has deficiency indices $(2M, 2M)$.

Proof. $T(k\mathbf{1} + cP_{2,M})$ is a polynomial in the single Hermitian matrix $P_{2,M}$, hence exactly diagonal in its constant (k -independent) eigenframe; the eigenvalues p_ν of $P_{2,M}$ (scaled Gauss–Hermite points) are simple. In that frame S_M is a direct sum of the scalar operators $\hat{p}^2/(2\gamma\Omega) - T(k + cp_\nu)$ —each limit circle at both ends [22], contributing $(2, 2)$ —plus the rotated channel-energy matrix, a constant bounded Hermitian perturbation, which preserves deficiency indices [23]. $\square$

The falsifiable consequence—with couplings on, the wall-regularized level density must equal the all- $2M$ -branch sum $\sum_\nu T_{\text{cross},\nu}/\pi$; were any channel’s deficiency destroyed, the density would drop—is confirmed numerically for $M = 1, 2, 3$ at three wall positions (count-based spacing matching the prediction within count granularity; the exact frame decoupling holds to 10^{-13} , and the $\pm p_\nu$ branch ladders visibly lock into parity doublets). What remains for the full operator is only the channel tail: the exact Fock compression differs from the Galerkin model by terms supported on the top four channels, and control of the deficiency vectors’ decay in channel number as $M \rightarrow \infty$ is open—equivalently, the full operator is a direct integral of limit-circle fibers over $\text{spec}(p_2)$ coupled across fibers by the transverse oscillator.

Corollary 9.9 (sign dichotomy at every finite channel number). *For $\lambda < 0$ the same frame decoupling yields the confining branches $\hat{p}^2/(2\gamma\Omega) + |\lambda|(k + cp_\nu)^4 + O(k^2)$, each essentially*

self-adjoint; with the bounded rotated channel-energy term (Kato–Rellich), every finite-channel Galerkin model of the $\lambda < 0$ quartic PU is essentially self-adjoint. Thus at every M the deficiency indices are $(2M, 2M)$ for $\lambda > 0$ and $(0, 0)$ for $\lambda < 0$.

Numerically, the $\lambda < 0$ wall-regularized spectra converge as the wall recedes (matched drift 3×10^{-8} , then 10^{-13})—the regulator-independence fingerprint, in maximal contrast to the never-converging $\lambda > 0$ sweep.

Mechanisms of the two completion candidates. Tracking the actual classical blowup trajectories and evaluating the kinetic form on the escaping momentum discriminates the mechanisms sharply. The $\lambda > 0$ PU (the deficiency case) escapes along a *definite* kinetic direction (form ratio $\rightarrow 0.9997$). The ghost pair’s escape is *null*: along its diagonal valley $p_1^2 - p_2^2$ stays below 0.07 of $p_1^2 + p_2^2$ —a direction supporting no definite-kinetic WKB transport, which is a structural rationale for its quantum completion. We conjectured that completion is equivalent to null escape; the $\lambda < 0$ PU *falsifies* the naive form of this conjecture: its ratio also tends to 1. Inspection shows it climbs a *rising* potential using negative kinetic energy while hugging the line $u = -cv$ —that is, it escapes through arbitrarily high mode-2 channels, precisely the channel tail excluded by the Galerkin models. The two candidates therefore realize two distinct completion mechanisms (null-direction escape versus channel-tail escape), the second more tail-sensitive than the first. Finally, the ghost-pair verdict is robust under valley-adapted bases: in rotated (s, d) Fock bases aligned with the escape diagonal, at three squeezing choices including strongly valley-penetrating ones, the $\lambda < 0$ drift ratios (0.16–0.45) interleave with the free ($\lambda = 0$) control’s range—nothing approaching the limit-circle band.

Finally, the $M \rightarrow \infty$ tail itself.

Proposition 9.10 (direct-integral deficiency; the exact form of the tail question). *The oscillator-free part $A = \hat{p}_k^2/(2\gamma\Omega) - T(k + cp_2)$ commutes with p_2 : it is a direct integral over $\text{spec}(p_2) = \mathbb{R}$ of shifted scalar fibers, each limit circle for $\lambda > 0$, and measurable fields of fiber deficiency solutions give $n_{\pm}(A) = \infty$. The full channelized operator differs from A by the transverse oscillator $h_2 = p^2/(2\gamma) - (\gamma\Omega/2)\partial_p^2$, which in this exact frame is unbounded (its Galerkin compressions have norm $\sqrt{\Omega}(M - \frac{1}{2}) \rightarrow \infty$)—precisely why the finite-channel stability theorem (Theorem 9.8) does not pass trivially to $M \rightarrow \infty$.*

Remark 9.11 (no compact witnesses). For any symmetric operator defined on C_0^∞ and any $\text{Im } z \neq 0$, every compactly supported smooth ψ lies in the closure’s domain, where $\|(S - z)\psi\| \geq |\text{Im } z| \|\psi\|$ holds unconditionally. *No compactly supported—hence no truncated, windowed, or lattice-regularized—function can witness deficiency*: the completion data lives entirely at infinity, in the same sense in which the completions themselves are an S -matrix at infinity (Theorem 9.2). All truncation-based diagnostics, including ours, are therefore evidence in principle, never proof; genuine von Neumann certificates require candidates with analytically controlled tails.

Such a certificate is implementable in one dimension: gluing WKB tails (whose only residual is the explicit remainder $a[q''/(4q) - \frac{5}{16}(q'/q)^2]$, $q = (z - U)/a$) to an exactly integrated interior yields, for the falling-quartic fiber, a machine-checked ratio $\|(S - iZ)\psi\|/(Z\|\psi\|) = 0.0011 \ll 1$: the 1D fiber operator is *certified*—in the computer-assisted sense: the residual is evaluated numerically with a three-orders safety margin, not rigorously error-bounded—not essentially self-adjoint. The construction is intrinsically sharp: for the confining control the left continuation contains a non- L^2 real-exponential branch (no candidate exists), and for the marginal falling-quadratic case the growing branch is polynomial, $|\psi| \sim k^{-1/2+Z/1.12}$, non- L^2 for every $Z > 0$ —consistent with the limit-point verdict of Theorem 8.1(1). On the full two-dimensional operator, windowed

candidates supply a *closure-defect diagnostic*: falling-channel operators hug the forbidden bound (ratios 1.21–1.41 for the five $\lambda > 0$ beam launches—four at 1.21–1.29, the outer-window launch at 1.41—bracketing the 1D limit-circle control at 1.22), while the $\lambda < 0$ control sits at 641 and the 1D confining control at 4.9. The remaining mathematical content of the $M \rightarrow \infty$ question is thereby isolated: a two-dimensional certificate requires analytic beam tails along the escape rays.

A Maslov feasibility study (`maslov_feasibility.py`) settles the geometry of those tails. In the null-frame coordinates the ray equations of the symbol reduce *exactly* to the quartic-PU law $s'''' + \Omega s'' + \Omega_0^2 s = (4\lambda/\gamma) s^3$, so the escape-ray family is the classical blowup itself, with universal asymptotics $s \simeq S(t_* - t)^{-2}$, $S = \sqrt{30\gamma/\lambda}$: measured exponent -1.999 and amplitude 24.55 against $\sqrt{600} = 24.49$, at every launch point tested, in *both* time directions—rest data escapes at both temporal ends, so the time-integral quasimode $\psi = \int e^{izt} \varphi_t dt$ over a doubly-explosive ray has only $e^{-Zt_*} \approx 10^{-14}$ boundary terms. Along these rays the Gaussian-beam data [49] are favorable precisely where analytic tails are needed: Siegel positivity of the complex curvature holds throughout, its numerical-zero direction aligned with the flow (alignment 1.0000; benign, integrated over in t); the transverse curvature stays positive with strong terminal focusing (width 5×10^{-3} at the escape end); no caustics occur; $|\det \partial q| \sim \tau^{-3.1}$ makes the beam-norm integral converge ($N = 0.082$ at $Z = 6$); and the anharmonic residual proxy decays rapidly along the escape (weighted integral 0.026), failing only in a *compact interior episode* of transverse defocusing (peak 8×10^3)—exactly the region the one-dimensional template already handles by exact numerical solution. The two-dimensional glued certificate thus reduces to: numeric interior on a compact region, Gaussian-beam tails along the known rays, gluing where the residual proxy is small—assembly and quantitative error control rather than new asymptotic machinery.

10 Experimental probes

The closed forms of this paper are testable with existing devices; we collect the concrete predictions. (*Added in v1.1.*)

(a) *The terminal-indivisibility theorem in a lossy photonic coupler.* The dimer (4) is the standard system of experimental $\mathcal{PT}$ optics (coupled waveguides or resonators with engineered loss), and the correspondence’s objects are normalized intensities, so classical light suffices. The apparatus exists: passive $\mathcal{PT}$ lossy couplers are characterized as functions of propagation length [40], information-flow criticality at the exceptional point has been measured in single-photon dimers [41], and divisibility *classes* (P- vs CP-divisibility) are experimentally resolvable [42]. What has not been extracted is precisely this paper’s observable set: a closed-form divisibility-onset length, a generator floor, and the recurrent-window period. Protocol: measure the intensity transfer matrix $\Gamma(t)$ at a sequence of propagation lengths, form the divisor $M = \Gamma(t_2)\Gamma(t_1)^{-1}$ (2), and locate the onset of negativity. Theorem 4.1 predicts, with no free parameters beyond the device constants (θ, s) : onset at $t_c = \tilde{\kappa}^{-1} \operatorname{arcsinh}(\tilde{\kappa}/(s\sqrt{2}))$, *permanent* thereafter in the broken phase, with the generator floor $-(\sin \theta + \tilde{\kappa})$; *recurrent* windows of period π/κ in the unbroken phase; and t_c continuous through the exceptional point. The dissociation of Sec. 4 is testable on the same data set by running the two analyses (trace-distance backflow versus divisibility) side by side: backflow dies at the exceptional point while the divisibility onset crosses it smoothly.

(b) *The conditioning signal on an entangled pair.* Applying the local lossy dimer to one arm of an entangled photon pair and post-selecting on survival shifts the partner’s marginal by a closed-form amount with late-time magnitude $\sqrt{1 - (s/\sin \theta)^2}$ —zero at the exceptional point, approaching unity deep in the broken phase—while the *unconditioned* marginal is exactly

unchanged (derivation in the companion sequel, available in the same repository; the closed form follows from Theorem 4.1’s p, q, r). This is a direct laboratory test of the sample-space bookkeeping that this paper argues is the content of broken- $\mathcal{PT}$ physics, and it locates the apparent no-signaling violation of local $\mathcal{PT}$ operations reported in the literature in the post-selection, quantitatively. The entangled-pair-plus-tailored-loss platform exists [43, 44]; the closed-form magnitude is the new target.

(c) *The skin obstruction in circuits and lattices.* Hatano–Nelson physics is routinely realized in topoelectrical circuits and photonic lattices; Theorem 3.4 identifies the measured skin envelope with the beable metric, $d_i = e^{2wi}$, and Corollary 3.5 predicts a specific new experiment: two opposite-field chains coupled by reciprocal rungs (the $\mathbb{Z}_2$ -type ladder, net winding zero) are obstructed at *any* rung coupling—observable as irreducible nonreciprocal transport that no local gauge transformation removes. The $\mathbb{Z}_2$ skin effect itself has recently been observed (acoustically and in programmable circuits [45]); the prediction here is not its existence but the *beable-obstruction observable*: the irreducibility of the nonreciprocity, equivalently the detailed-balance violation of the underlying rate graph, which mode-localization measurements do not extract.

(d) *The explosion dictionary in engineered potentials.* Level-spacing *saturation*, $\Delta E = \pi/T_{\text{cross}}$ independent of the box size (Table 1), is the spectroscopic fingerprint of reflection off infinity at finite recurrence time (the underlying boundary-condition spectra are classical mathematical physics [46]; the saturation framing as an observable appears new); engineered super-quadratically falling potentials in cold-atom or circuit-QED simulators could exhibit it directly—inverted-oscillator BEC platforms with full state tomography now exist [47] and are one nonlinearity away from the $-x^4$ class. On the transient side, the cascade laws (14) are two-mode squeezing, already measured as vacuum pair creation in the dynamical Casimir effect [48]; engineered superlinearly Bose-enhanced ladders would probe the transient-to-explosive transition itself.

11 Discussion

11.1 What each program learns

For the antilinearity program. Closed-system $\mathcal{PT}$ machinery adds no new stochastic processes: in the unbroken phase it is an isomorphism onto ordinary quantum mechanics (the deflation proposition, Proposition 2.1), at the EP its coordinates degenerate (Sec. 5), and its genuinely new regime—the broken phase—is most naturally read as *open-system* physics: survival-conditioned dynamics whose stochastic signature is the recurrent $\rightarrow$ terminal transition of indivisibility (Theorem 4.1). For PU, the no-ghost construction is domain data at infinity rather than dynamics on the configuration space (Sec. 6); the explosion dictionary (Theorem 8.1(2)(a)) and the channel-additivity theorem (Theorem 9.2) show precisely what such data amounts to.

For the stochastic–quantum correspondence. Antilinear symmetry is the exact existence gate for closed processes; detailed balance (the detailed-balance theorem, Theorem 3.1) is the exact fixed-beable gate, resolving both conceptual costs on that class and giving the metric weights a classical meaning (inverse stationary distribution). The correspondence handles ghosts gracefully—too gracefully to see them at free level: the ghost enters the stochastic currency only through recurrence classification (conservative $\rightarrow$ transient $\rightarrow$ explosive), and the minimal explosive process is the only completion-independent object.

11.2 The trilemma

For a malicious interacting ghost one must choose: (i) accept the minimal explosive process—sub-stochastic, the dilation structure of Sec. 4, canonical; (ii) buy conservativity with an S -matrix at infinity—in multi-D an infinite-dimensional, channel-scrambling reflection law injected by hand (Theorem 9.2); or (iii) repair the spectrum by moving the domain—which exits the sample space (Sec. 6). In Barandes’ currency, (i) is the answer; vacuum decay *is* explosion in the probabilist’s standard sense.

11.3 Outlook

Open directions: Open Problem 9.6, whose two-dimensional certificate is now reduced to assembly and error control over the beam data supplied in Sec. 9; field theory, where the explosion/transience distinction should meet vacuum decay rates; the interacting broken phase, where terminal indivisibility and transience could combine; and extensions of the skin-effect correspondence of Sec. 3.1 to higher-dimensional and symmetry-protected skin effects [38], where the obstruction should acquire the corresponding topological classification.

Author contributions and use of AI

This manuscript was produced through a collaboration, directed by the author, with the large language model Claude Fable 5 (Anthropic). The AI system carried out essentially all of the technical work: the research program and its strategy, the analytical derivations and proofs, the design, implementation, and interpretation of all numerical experiments and figures, the literature search and reference verification, and the drafting and revision of the manuscript, including this statement. J.T. posed the research direction, provided iterative review and quality feedback, and takes sole responsibility for the decision to circulate the manuscript. In line with prevailing editorial policies (e.g., COPE guidance and the policies of the major physics publishers), the AI system is not listed as an author, since authorship entails accountability that an AI system cannot hold; this statement is intended to make the provenance unambiguous. Every numerical claim is reproducible from the accompanying scripts, and every theorem-level claim is presented with its proof, so that all results can be verified independently of their origin.

Code availability

All scripts, the running lab notes, and the L^AT_EX source of this manuscript are publicly available at

<https://github.com/embeddetech/p01-indivisible-ghosts>

The repository also preserves the provenance record: the original conversation in which the research question was posed (`Conversation_Conformal_Gravity_PT_Barandes.pdf`) and the project briefing that seeded the working sessions (`PT_stochastic_handoff.md`). All results are reproducible from short standalone Python scripts (NumPy/SciPy); most run in seconds to minutes on a laptop, while the two long trajectory-integration scripts (`interacting_pu.py`, `completion_candidates.py`) can take tens of minutes, machine- and SciPy-version-dependent:

- Sec. 2: `pt_barandes.py`

- Sec. 3: `fixed_beable_kolmogorov.py`, `skin_effect_beables.py`, `skin_topology_beables.py`
- Sec. 4: `dilation_bridge.py`, `lemma_terminal_indivisibility.py`
- Sec. 5: `exceptional_point.py`
- Sec. 6: `pais_uhlenbeck.py`
- Sec. 7: `interacting_pu.py`, `quartic_pu_leak.py`
- Sec. 8: `explosion_theorem.py`
- Sec. 9: `deficiency_multiD.py`, `pu_deficiency_evidence.py`, `pu_bo_channels.py`, `gap_closure_channels.py`, `completion_candidates.py`, `tail_certificate.py`, `glued_certificate_id.py`, `maslov_feasibility.py`
- Figures: `paper/make_figures.py` regenerates all figures except Fig. 1, which is produced by `skin_effect_beables.py`

References

- [1] J. A. Barandes, *The stochastic-quantum correspondence*, Philos. Phys. **3**(1), 8 (2025), arXiv:2302.10778.
- [2] J. A. Barandes, *Quantum systems as indivisible stochastic processes*, arXiv:2507.21192.
- [3] L. S. Pimenta, *Divisible and indivisible Stochastic-Quantum dynamics*, arXiv:2505.08785.
- [4] C. M. Bender and S. Boettcher, *Real spectra in non-Hermitian Hamiltonians having $\mathcal{PT}$ symmetry*, Phys. Rev. Lett. **80**, 5243 (1998).
- [5] C. M. Bender, *Making sense of non-Hermitian Hamiltonians*, Rep. Prog. Phys. **70**, 947 (2007).
- [6] P. D. Mannheim, *Antilinearity rather than Hermiticity as a guiding principle for quantum theory*, J. Phys. A **51**, 315302 (2018), arXiv:1512.04915.
- [7] P. D. Mannheim, *Making the case for conformal gravity*, Found. Phys. **42**, 388 (2012).
- [8] C. M. Bender and P. D. Mannheim, *No-ghost theorem for the fourth-order derivative Pais–Uhlenbeck oscillator model*, Phys. Rev. Lett. **100**, 110402 (2008).
- [9] A. Pais and G. E. Uhlenbeck, *On field theories with non-localized action*, Phys. Rev. **79**, 145 (1950).
- [10] A. Mostafazadeh, *Pseudo-Hermiticity versus $\mathcal{PT}$ symmetry: The necessary condition for the reality of the spectrum of a non-Hermitian Hamiltonian*, J. Math. Phys. **43**, 205 (2002); *Pseudo-Hermitian representation of quantum mechanics*, Int. J. Geom. Meth. Mod. Phys. **7**, 1191 (2010).
- [11] A. V. Smilga, *Benign vs. malicious ghosts in higher-derivative theories*, Nucl. Phys. B **706**, 598 (2005).

- [12] P. R. Halmos, *Normal dilations and extensions of operators*, Summa Brasil. Math. **2**, 125 (1950).
- [13] B. Sz.-Nagy, C. Foias, H. Bercovici, and L. Kérchy, *Harmonic Analysis of Operators on Hilbert Space*, 2nd ed. (Springer, New York, 2010).
- [14] H.-P. Breuer, E.-M. Laine, and J. Piilo, *Measure for the degree of non-Markovian behavior of quantum processes in open systems*, Phys. Rev. Lett. **103**, 210401 (2009).
- [15] Á. Rivas, S. F. Huelga, and M. B. Plenio, *Entanglement and non-Markovianity of quantum evolutions*, Phys. Rev. Lett. **105**, 050403 (2010).
- [16] H.-P. Breuer, E.-M. Laine, J. Piilo, and B. Vacchini, *Colloquium: Non-Markovian dynamics in open quantum systems*, Rev. Mod. Phys. **88**, 021002 (2016).
- [17] A. N. Kolmogorov, *Zur Theorie der Markoffschen Ketten*, Math. Ann. **112**, 155 (1936).
- [18] F. P. Kelly, *Reversibility and Stochastic Networks* (Wiley, Chichester, 1979).
- [19] N. Hatano and D. R. Nelson, *Localization transitions in non-Hermitian quantum mechanics*, Phys. Rev. Lett. **77**, 570 (1996).
- [20] K. Petermann, *Calculated spontaneous emission factor for double-heterostructure injection lasers with gain-induced waveguiding*, IEEE J. Quantum Electron. **15**, 566 (1979).
- [21] H. Weyl, *Über gewöhnliche Differentialgleichungen mit Singularitäten und die zugehörigen Entwicklungen willkürlicher Funktionen*, Math. Ann. **68**, 220 (1910).
- [22] M. Reed and B. Simon, *Methods of Modern Mathematical Physics II: Fourier Analysis, Self-Adjointness* (Academic Press, New York, 1975), Thms. X.8–X.10.
- [23] T. Kato, *Perturbation Theory for Linear Operators*, 2nd ed. (Springer, Berlin, 1976), stability of deficiency indices under relatively bounded symmetric perturbations.
- [24] E. Nelson, *Analytic vectors*, Ann. Math. **70**, 572 (1959).
- [25] W. Feller, *The parabolic differential equations and the associated semi-groups of transformations*, Ann. Math. **55**, 468 (1952).
- [26] J. Rauch and M. Reed, *Two examples illustrating the differences between classical and quantum mechanics*, Commun. Math. Phys. **29**, 105 (1973).
- [27] S. V. Parter and J. W. T. Youngs, *The symmetrization of matrices by diagonal matrices*, J. Math. Anal. Appl. **4**, 102 (1962).
- [28] G. M. Engel and H. Schneider, *Cyclic and diagonal products on a matrix*, Linear Algebra Appl. **7**, 301 (1973).
- [29] T. Van Wesemael, G. Nakamura, J. Baetens, O. M. Bruno, A. S. Martinez, and C. Deroulers, *Restoring detailed balance in non-Hermitian Markov processes*, arXiv:2510.09467.
- [30] K. Kawabata, Y. Ashida, and M. Ueda, *Information retrieval and criticality in parity-time-symmetric systems*, Phys. Rev. Lett. **119**, 190401 (2017).

- [31] D. Chruściński, Á. Rivas, and E. Størmer, *Divisibility and information flow notions of quantum Markovianity for noninvertible dynamical maps*, Phys. Rev. Lett. **121**, 080407 (2018).
- [32] Á. Rivas, *Unidirectional information flow and positive divisibility are nonequivalent notions of quantum Markovianity for noninvertible dynamics*, Open Syst. Inf. Dyn. **29**, 2250012 (2022), arXiv:2209.03584.
- [33] A. Salvio and A. Strumia, *Quantum mechanics of 4-derivative theories*, Eur. Phys. J. C **76**, 227 (2016).
- [34] V. Errasti Díez, J. Gaset Rifà, and G. Staudt, *Foundations of ghost stability*, Fortschr. Phys. **73**, 2400268 (2025), arXiv:2408.16832.
- [35] A. Held, *Global stability of ghostly field theories: Classical scattering in $(N+1)$ dimensions*, arXiv:2509.18049.
- [36] C. Deffayet, A. Fathe Jalali, A. Held, S. Mukohyama, and A. Vikman, *Unitary time evolution and vacuum for a quantum stable ghost*, arXiv:2604.21823.
- [37] C. Ewasiuk and S. Profumo, *Ghost degrees of freedom without quantum runaway: exact moment bounds from an operator conservation law*, arXiv:2604.21348.
- [38] N. Okuma, K. Kawabata, K. Shiozaki, and M. Sato, *Topological origin of non-Hermitian skin effects*, Phys. Rev. Lett. **124**, 086801 (2020).
- [39] F. Fischer, D. Burgarth, and D. Lonigro, *Self-adjoint realizations of higher-order squeezing operators*, J. Phys. A **59**, 255203 (2026), arXiv:2508.09044.
- [40] F. Klauck, L. Teuber, M. Ornigotti, M. Heinrich, S. Scheel, and A. Szameit, *Observation of $\mathcal{PT}$ -symmetric quantum interference*, Nat. Photonics **13**, 883 (2019).
- [41] L. Xiao, K. Wang, X. Zhan, Z. Bian, K. Kawabata, M. Ueda, W. Yi, and P. Xue, *Observation of critical phenomena in parity-time-symmetric quantum dynamics*, Phys. Rev. Lett. **123**, 230401 (2019).
- [42] N. K. Bernardes et al., *Experimental observation of weak non-Markovianity*, Sci. Rep. **5**, 17520 (2015).
- [43] J.-S. Tang et al., *Experimental investigation of the no-signalling principle in parity-time symmetric theory using an open quantum system*, Nat. Photonics **10**, 642 (2016).
- [44] M. Ehrhardt, M. Heinrich, and A. Szameit, *Observation-dependent suppression and enhancement of two-photon coincidences by tailored losses*, Nat. Photonics **16**, 191 (2022).
- [45] T. Liu et al., *Observation of $\mathbb{Z}_2$ non-Hermitian skin effect in projective mirror-symmetric acoustic metamaterials*, Adv. Mater. (2025); H. Tang et al., *A non-Abelian route to $\mathbb{Z}_2$ non-Hermitian skin effects*, arXiv:2604.00888.
- [46] M. Carreau, E. Farhi, S. Gutmann, and P. F. Mende, *The functional integral for quantum systems with Hamiltonians unbounded from below*, Ann. Phys. **204**, 186 (1990).

- [47] S.-C. Ji, P. Schüttelkopf, N. Bazhan, F. Cataldini, M. Tajik, F. S. Møller, I. Mazets, S. Erne, and J. Schmiedmayer, *Experimentally probing the quantum physics in the inverted harmonic oscillator*, arXiv:2606.05125.
- [48] C. M. Wilson et al., *Observation of the dynamical Casimir effect in a superconducting circuit*, Nature **479**, 376 (2011).
- [49] J. Ralston, *Gaussian beams and the propagation of singularities*, in *Studies in Partial Differential Equations*, MAA Studies in Mathematics **23**, 206–248 (Math. Assoc. America, Washington, DC, 1982).